Recursive Bayesian Compression and False Certainty*

Jonghyun Park†

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Abstract

A fully Bayesian learner carries forward the complete joint posterior over a state with two components. We study a learner who instead stores only the posterior distribution of each component separately. Before each new signal, she reconstructs a joint belief by treating the two stored distributions as independent, applies Bayes' rule to that reconstructed belief, and again stores only the resulting marginals. Thus every signal is observed and every update is Bayesian, but the relationship between the two state components created by earlier observations is discarded before later evidence is processed. We show that this recursive use of marginal records can produce false certainty. We show the possible convergence set under this recursive learning. The learner can converge to a state that fits the true signal distribution better than every alternative obtained by changing only one of its components, even though another state fits the signal distribution better overall. We also characterize exactly when marginal records lose no information needed for future learning. Starting from any prior, recursive marginal learning reproduces full Bayesian learning after every finite signal history if and only if, for each signal realization, its likelihood is the product of one positive term depending only on the first component and another depending only on the second.

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†Department of Economics, Virginia Tech

1 Introduction

Economic learning often concerns several related dimensions of an uncertain state. A loan officer may assess both a borrower's underlying quality and whether recent losses reflect growth or solvency risk. A hiring committee may assess both ability and organizational fit. A manager may receive separate measures of several risks rather than the evidence from which those assessments were formed. After each new observation, a fully Bayesian learner can retain the complete history or, equivalently for future Bayesian updating, the full joint belief that summarizes everything the history implies about the different dimensions, including their posterior association.

Actual records often retain less. Rather than storing the full joint belief, an organization may record only its assessment of each dimension separately: one probability for underlying quality and another for the source of recent losses, or one score for ability and another for organizational fit. We call these *marginal records*. They preserve the current assessment of each state component but not the posterior association between the components. Different joint beliefs can therefore generate exactly the same marginal record.

This omission may be harmless when the record is used only to report current assessments. It can matter when the same record is later used to interpret new evidence. Suppose that an organization records the probability that a project is strong and the probability that recent losses reflect growth rather than solvency problems. These two probabilities do not reveal whether strength is believed to be especially likely when losses reflect growth. Yet that omitted association can affect how the next observation should change both assessments. A record can therefore report each marginal assessment exactly while still omitting information needed for future learning.[1]

We study this problem in an environment with a state that has two components. The learner observes every signal, knows the correct likelihood system, and applies Bayes' rule. Her only restriction concerns what she carries from one date to the next. After processing a signal, she retains the posterior distribution of each component separately but not their posterior association. Before processing the next signal, she reconstructs a joint belief by treating the two recorded marginals as independent. She applies Bayes' rule to this reconstructed product belief and again stores only the resulting marginals. We call this procedure *recursive marginal learning*.

The record produced at each date is exact in a specific sense: it contains the marginals of the within-period posterior formed from the reconstructed product belief. It need not equal

[1]We develop and discuss this example in Section 3.

the marginal report of the posterior that a full Bayesian learner would form from the entire signal history. The restriction is therefore intertemporal. A correct Bayesian update can create posterior association, and that association can disappear from the record before the next signal is processed.

This form of compression captures a common feature of economic records. Organizations often transmit scorecards, files, dashboards, standardized evaluations, or short memoranda rather than complete posterior distributions or case histories. Experimental evidence suggests that people may underuse relationships among pieces of information or infer joint relationships from separate frequencies (Enke and Zimmermann, 2019; Fiedler, 2010). High-stakes decisions are also increasingly mediated by prediction scores, risk assessments, and algorithmic recommendations (Angelova et al., 2026; Kleinberg et al., 2018; Stevenson and Doleac, 2024). These settings motivate the economic object studied here: a compact record that preserves separate assessments while omitting their posterior association.

Our first result characterizes the pure states that recursive marginal learning can select. Each coordinate is evaluated using the current marginal assessment of the other coordinate. A state can therefore be selected only if it explains the data at least as well as every alternative obtained by changing one coordinate while holding the other fixed. We call such a state a weak coordinate-wise KL minimum. Conversely, every strict coordinate-wise KL minimum attracts records that sufficiently favor both of its components, with the probability of convergence approaching one as that initial support increases.

The distinction between coordinate-wise and global fit permits false certainty. A false state may fit the true signal distribution worse than another state when the two joint states are compared directly, while still fitting the data better than every alternative obtained by changing only one of its coordinates. Once the marginal record strongly favors both components of the false state, revising either assessment alone appears inconsistent with the evidence. Correcting the error requires revising both coordinates together, but the posterior association needed to preserve that joint comparison is precisely what the marginal record has discarded.

A binary example makes the mechanism transparent. A project can be strong or weak, and its losses can reflect either growth or solvency risk. The true state is that the project is weak and faces solvency risk. There is nevertheless a false state in which the project is strong and its losses reflect growth. The true state explains the signal distribution better overall. But if the learner maintains the growth assessment, the evidence favors a strong project over a weak one; if she maintains the strong-project assessment, the evidence favors growth over solvency risk. The two marginal assessments therefore validate one another. A learner who retains the full joint posterior eventually rejects the false interpretation, whereas a learner who recursively reuses only the two marginals can become certain of it.

The failure is neither confined to records initially close to the false state nor tied to the exact numerical example. Throughout an open set of correctly specified likelihood systems in which the true state is identified, the full Bayesian learner learns the truth from every full-support prior. From every full-support initial record, the recursive marginal learner instead converges almost surely to either the true or the false diagonal state, and each limit occurs with positive probability. The two learners can therefore begin from the same product belief, observe the same realized signal history, and know the same likelihood system, yet reach different long-run conclusions solely because one carries posterior association forward and the other repeatedly deletes it.

We also identify the exact boundary at which marginal records lose nothing needed for future learning. Recursive marginal learning reproduces full Bayesian learning from every product prior after every finite signal history if and only if, for each signal realization, its likelihood factors into one positive term depending only on the first state coordinate and another depending only on the second. Under this no-coordinate-interaction condition, Bayesian updating from a product belief never creates posterior association, so the complete posterior remains determined by its two marginals. If both state coordinates are binary and this condition fails, the omitted information has one dimension: retaining the two marginals together with one additional posterior association coordinate is sufficient to reconstruct the full posterior exactly.

The broader implication is that the quality of an informational record is inherently dynamic. A record should be evaluated not only by whether it accurately reports current assessments, but also by whether it preserves the relationships needed to interpret future evidence.

1.1 Related literature

The closest paper is Loh and Phelan (2019), which studies a related procedure in which agents retain marginal beliefs and reconstruct joint beliefs by imposing independence across dimensions. They show that agents who observe common information can interpret it differently, sustain disagreement, and converge to false limiting beliefs. We take this mechanism as a starting point but focus on different characterization questions. We identify which pure states can be selected, establish robust same-history false certainty in an identified binary environment, and characterize exactly when marginal records reproduce full Bayesian learning from every product prior.

The paper is also related to Bayesian learning under misspecification. Standard results show that when the learner's model may exclude the truth, posterior beliefs favor the models that best approximate the true data-generating process (Berk, 1966; Bunke and Milhaud, 1998). This logic has been used extensively in economic models of learning and long-run behavior under subjective models (Esponda and Pouzo, 2016; Esponda et al., 2021). Our source of error is different. In the false-certainty environments studied below, the true state belongs to the learner's model, the likelihoods are correct, every signal is observed, and the true state is identified, so unrestricted Bayesian learning converges to the truth. The distortion arises because part of a correctly formed posterior is deleted before later observations are processed.

A related literature studies simplified representations and limited attention. Agents may group rich situations into coarse categories, attend selectively to predictive variables, or fail to encode information needed for later learning (Gennaioli and Shleifer, 2010; Hanna et al., 2014; Mullainathan et al., 2008; Schwartzstein, 2014). Our learner retains the current marginal belief about every state component. What is omitted is the posterior association between those components.

The mechanism is also connected to work on neglected dependence. Experimental evidence documents underreaction to correlation and reliance on marginal frequencies when joint relationships matter (Eder et al., 2011; Enke and Zimmermann, 2019; Fiedler, 2010; Fiedler and Freytag, 2004). Related theoretical work studies failures to account for dependence between actions and information, redundancy in social learning, and omitted links in subjective representations (Angrisani et al., 2021; Eyster and Rabin, 2005; Eyster et al., 2018; Spiegler, 2016). In our model, the learner knows the objective dependence encoded in the likelihood system. The relevant association instead arises endogenously in the posterior through correct Bayesian updating and is then removed from the record.

The rest of the paper proceeds as follows. Section 2 defines marginal records and the two learning procedures. Section 3 presents the binary example. Section 4 characterizes pure-state selection and reinforcement, and Section 5 studies their global consequences in binary environments. Section 6 identifies when marginal records preserve everything needed for future Bayesian updating and gives an exact binary augmentation when they do not. Section 7 concludes.

2 Marginal Records and Recursive Learning

The state has two components, and the learner observes an infinite sequence of signals about their joint realization. We compare two learning procedures that observe the same signals, know the same correctly specified likelihood system, and use the same Bayesian update operator. They differ only in the information they carry from one date to the next. A full Bayesian learner retains the joint posterior. A recursive marginal learner retains only its two marginals and reconstructs their product before processing the next signal. This reconstruction preserves the current assessment of each component separately but removes the posterior association between the state coordinates.

2.1 Environment and learning procedures

For a finite set S, let $\Delta(S)$ denote the set of full-support probability distributions on S, and let $\overline{\Delta}(S)$ denote its closure, the set of all probability distributions on S.[2] For any finite set E and any $e \in E$, let $\delta_e \in \overline{\Delta}(E)$ denote the degenerate distribution concentrated on e.

Let A, B, and Y be finite sets, with $|A|, |B| \geq 2$, and let the state be $x = (a, b) \in X := A \times B$. Conditional on state (a, b), the signals $(Y_t)_{t \geq 1}$ are independently and identically distributed according to $P_{ab} \in \Delta(Y)$. The true state is $x^* = (a^*, b^*)$, and the true signal distribution is $P^* := P_{a^*b^*}$. Write $\mathbb{P}^*$ and $\mathbb{E}^*$ for probability and expectation under the true state. Every learner knows the likelihood system $\{P_{ab} : (a, b) \in X\}$, so the statistical model is correctly specified.

For any finite set E and any $\mu, \nu \in \overline{\Delta}(E)$, define

$$D_{\mathrm{KL}}(\mu \| \nu) := \sum_{e \in E} \mu(e) \log \frac{\mu(e)}{\nu(e)},$$

with the convention that terms with $\mu(e) = 0$ are zero and that $D_{\mathrm{KL}}(\mu \| \nu) = +\infty$ if $\nu(e) = 0 < \mu(e)$ for some $e \in E$. The KL divergence between the true signal distribution and the likelihood associated with state (a, b) is

$$K(a, b) := D_{\mathrm{KL}}(P^* \| P_{ab}) = \sum_{y \in Y} P^*(y) \log \frac{P^*(y)}{P_{ab}(y)}.$$

For $x = (a, b)$, we also write $K(x) := K(a, b)$. Thus $K(a^*, b^*) = 0$. We say that the true

[2] We reserve the unbarred notation for likelihoods, priors, and finite-date beliefs, all of which have full support. Full support keeps likelihood ratios finite and is preserved by Bayesian updating. The closure $\overline{\Delta}(S)$ is used for degenerate limiting beliefs and for statements extended to the boundary.

state is *identified* if $K(a,b) > 0$ for every $(a,b) \neq (a^*, b^*)$.[3]

Let

$$Q := \Delta(A) \times \Delta(B), \qquad \overline{Q} := \overline{\Delta}(A) \times \overline{\Delta}(B).$$

A finite-date marginal record is a pair $q = (\alpha, \beta) \in Q$. The closed record space $\overline{Q}$ also contains the boundary records used to describe limiting beliefs. A record of the form (δ_a, δ_b) is the *pure record* associated with state $x = (a,b)$.

Given a joint belief $\pi \in \overline{\Delta}(X)$, define its marginal record by

$$\mathsf{M}(\pi) := (\pi_A, \pi_B), \qquad \pi_A(a) := \sum_{b \in B} \pi(a,b), \qquad \pi_B(b) := \sum_{a \in A} \pi(a,b).$$

Given $q = (\alpha, \beta) \in \overline{Q}$, define its product reconstruction $R(q) \in \overline{\Delta}(X)$ by

$$R(q)(a,b) := \alpha(a)\beta(b).$$

A joint belief $\pi \in \overline{\Delta}(X)$ is a *product belief* if $\pi = R(q)$ for some $q \in \overline{Q}$, or equivalently if $\pi(a,b) = \pi_A(a)\pi_B(b)$ for every $a \in A$ and $b \in B$. Hence

$$\mathsf{M}(R(q)) = q, \qquad R(\mathsf{M}(\pi))(a,b) = \pi_A(a)\pi_B(b).$$

The composition $R \circ \mathsf{M}$ preserves both marginals of a joint belief while removing the association between the state coordinates under that belief.

Given $\pi \in \overline{\Delta}(X)$ and $y \in Y$, define the Bayesian update operator

$$\mathsf{B}_y(\pi)(a,b) := \frac{\pi(a,b)P_{ab}(y)}{\sum_{a' \in A}\sum_{b' \in B} \pi(a',b')P_{a'b'}(y)}.$$

Because every likelihood has full support, B_y is well defined on $\overline{\Delta}(X)$ and maps $\Delta(X)$ into itself. Define the one-period recursive update map by

$$U_y(q) := \mathsf{M}\big(\mathsf{B}_y(R(q))\big), \qquad q \in \overline{Q}.$$

The map U_y sends Q into Q.

[3] Note $(a,b) \neq (c,d)$ if $a \neq c$ or $b \neq d$.

Full Bayesian learning. A full Bayesian learner retains a joint posterior $\pi_t^F \in \Delta(X)$. Starting from $\pi_0^F \in \Delta(X)$, she updates according to

$$\pi_{t+1}^F := \mathsf{B}_{Y_{t+1}}(\pi_t^F). \tag{1}$$

For any $(a, b), (a', b') \in X$, iterating (1) gives

$$\log \frac{\pi_t^F(a, b)}{\pi_t^F(a', b')} = \log \frac{\pi_0^F(a, b)}{\pi_0^F(a', b')} + \sum_{\tau=1}^{t} \log \frac{P_{ab}(Y_\tau)}{P_{a'b'}(Y_\tau)}. \tag{2}$$

The full posterior therefore carries forward every pairwise comparison among joint states generated by the observed history.

Recursive marginal learning. Starting from $q_0 \in Q$, the recursive marginal learner carries forward a record $q_t = (\alpha_t, \beta_t) \in Q$ satisfying

$$q_{t+1} := U_{Y_{t+1}}(q_t) = \mathsf{M}\big(\mathsf{B}_{Y_{t+1}}(R(q_t))\big). \tag{3}$$

At date $t + 1$, she reconstructs the product prior $R(q_t)$, observes Y_{t+1}, applies Bayes' rule, and stores the two marginals of the resulting joint posterior. Thus q_{t+1} exactly records the marginals of the within-period posterior $\mathsf{B}_{Y_{t+1}}(R(q_t))$. The joint posterior formed within the date is not retained.

The recursive marginal learner therefore observes every signal and applies Bayes' rule to the reconstructed prior available at that date. Her restriction is intertemporal: posterior association generated by one update is absent from the prior used to interpret the next signal. The record can consequently be exact about the two marginal assessments it contains while omitting information relevant for subsequent updating.

Whenever the two procedures are compared, fix $q_0 \in Q$ and initialize the full Bayesian learner from the same product belief,

$$\pi_0^F := R(q_0).$$

The two dynamic objects are then the full-history posterior π_t^F, generated by (1), and the recursively reused record q_t, generated by (3). The object $\mathsf{M}(\pi_t^F)$ is the marginal report of the full Bayesian posterior after the history through date t has been processed. It is not used to process the next signal.

Although $q_0 = \mathsf{M}(\pi_0^F)$ for the comparison, the equality $q_t = \mathsf{M}(\pi_t^F)$ need not persist. Full Bayesian learning carries the posterior association contained in π_t^F into the next update.

Recursive marginal learning instead feeds q_t back into the process through $R(q_t)$, imposing independence before each subsequent signal is interpreted.

2.2 Static KL projection and the full Bayesian benchmark

We first show that the product reconstruction used by the recursive marginal learner is not an arbitrary approximation. At any fixed date, the product of the two marginals is the unique product representation that minimizes $D_{\mathrm{KL}}(\pi \| R(q))$.

Proposition 1. *For every $\pi \in \Delta(X)$, the marginal record $\mathsf{M}(\pi)$ is the unique solution to*

$$\min_{q \in \overline{Q}} D_{\mathrm{KL}}\big(\pi \| R(q)\big).$$

Proposition 1 establishes the static quality of the representation. Given any joint belief, the learner stores the marginal record whose product reconstruction minimizes $D_{\mathrm{KL}}(\pi \| R(q))$. The substantive question is therefore not whether the learner uses an arbitrary product approximation at a given date. It is whether this representation preserves the information needed when it is repeatedly reused to interpret subsequent signals.

The next proposition records the full Bayesian benchmark.

Proposition 2. *Let $(\pi_t^F)_{t \geq 0}$ be generated from any $\pi_0^F \in \Delta(X)$. If the true state is identified, then*

$$\pi_t^F \to \delta_{(a^*, b^*)}, \qquad \mathsf{M}(\pi_t^F) \to (\delta_{a^*}, \delta_{b^*}) \qquad \mathbb{P}^*\text{-}a.s.$$

The first conclusion is the standard finite-state Bayesian learning result. The second shows that marginal reporting is asymptotically harmless when the marginals continue to be computed from the full-history posterior. The problem is therefore not reporting the posterior marginals, but recursively using a marginal report as the prior state for future inference.

When both learners begin from $\pi_0^F = R(q_0)$, they have the same initial belief, observe the same realized signals, and use the same correctly specified likelihood system. Any divergence between q_t and $\mathsf{M}(\pi_t^F)$ must therefore result from replacing the joint posterior with the product of its marginals before the next signal is processed. It cannot be attributed to an incorrect model, unobserved signals, non-Bayesian within-period updating, or an arbitrary static approximation.

3 A Binary False-Certainty Example

This section presents a minimal binary environment in which the two stored marginals reinforce one another and draw the recursive marginal learner toward a false state. The example isolates the distinction between global fit and the coordinate-wise comparisons generated by recursive marginal learning.

Let

$$A = \{H, L\}, \qquad B = \{g, s\}, \qquad Y = \{0, 1\}.$$

The first coordinate describes project type, where H denotes a strong project and L a weak project. The second coordinate describes the diagnostic regime, where g denotes a growth regime and s a solvency-risk regime. The realization $y = 1$ is diagnostic and need not represent project success. Conditional on state (a, b), the signal is Bernoulli, with

$P_{ab}(1)$	g	s
H	$3/5$	$1/10$
L	$9/10$	$1/2$

and $\qquad P_{ab}(0) = 1 - P_{ab}(1).$

The true state is

$$x^* = (L, s), \qquad P^* = P_{Ls} = \text{Bernoulli}(1/2).$$

The four signal distributions are distinct, so the true state is identified. Proposition 2 therefore implies that, from any full-support prior,

$$\pi_t^F \to \delta_{(L,s)}, \qquad \mathsf{M}(\pi_t^F) \to (\delta_L, \delta_s) \qquad \mathbb{P}^*\text{-a.s.}$$

The KL divergences from the true signal distribution are

$$K(L, s) = 0, \qquad K(H, g) = \frac{1}{2} \log \frac{25}{24}, \qquad K(H, s) = K(L, g) = \frac{1}{2} \log \frac{25}{9}. \tag{4}$$

Thus (L, s) is the unique global KL minimizer. Nevertheless, the false state $\bar{x} = (H, g)$ fits the true signal distribution better than either state obtained by changing only one of its coordinates:

$$K(H, g) < K(L, g), \qquad K(H, g) < K(H, s). \tag{5}$$

Thus, holding either coordinate at its value in (H, g), the evidence favors retaining the other coordinate. The diagonal alternative (L, s) fits the true signal distribution perfectly, but reaching it from (H, g) requires changing both coordinates together.

To see how the inequalities in (5) affect the recursive dynamics, write $q_t = (\alpha_t, \beta_t)$. Consider first the learner's assessment of project type. Suppose that the recursive learner has a marginal assessment β_t on the diagnostic regime B. Conditional on β_t, the likelihood assigned to signal y under H is

$$\beta_t(g)P_{Hg}(y) + \beta_t(s)P_{Hs}(y),$$

while the corresponding likelihood under L is

$$\beta_t(g)P_{Lg}(y) + \beta_t(s)P_{Ls}(y).$$

The type update therefore compares H with L after averaging over the regime using the learner's current marginal assessment β_t. As $\beta_t(g)$ approaches one, these two effective likelihoods approach P_{Hg} and P_{Lg}, respectively. The expected value of the log odds of H against L therefore converges to[4]

$$\sum_{y\in Y} P^*(y)\log\frac{P_{Hg}(y)}{P_{Lg}(y)}.$$

The regime update has the same structure under α_t.

At the pure record (δ_H, δ_g), these limiting expected changes are

$$\sum_{y\in Y} P^*(y)\log\frac{P_{Hg}(y)}{P_{Lg}(y)} = K(L,g) - K(H,g) = \frac{1}{2}\log\frac{8}{3} > 0,$$
$$\sum_{y\in Y} P^*(y)\log\frac{P_{Hg}(y)}{P_{Hs}(y)} = K(H,s) - K(H,g) = \frac{1}{2}\log\frac{8}{3} > 0.$$

The first line says that, when the marginal assessment of the regime places probability near one on g, subsequent evidence tends on average to increase the log odds of H against L. The second says that, when the marginal assessment of project type places probability near one on H, subsequent evidence tends on average to increase the log odds of g against s. Because these expected log-odds changes vary continuously with the current marginal record, both remain strictly positive throughout a neighborhood of (δ_H, δ_g).

The two marginal assessments therefore reinforce one another. A high probability on g makes the expected type update favor H, while a high probability on H makes the expected regime update favor g. Neither assessment is reinforced in isolation: each changes the likelihood comparison used to update the other. This is why the false interpretation (H, g) can become locally self-reinforcing even though the true diagonal state (L, s) has strictly better global KL fit.

[4]Note that the signal y is generated under P^*.

A full Bayesian learner instead retains the direct posterior comparison between the true diagonal state (L, s) and the false diagonal state (H, g). Applying the posterior log-odds identity (2) and using (4) gives

$$\frac{1}{t} \log \frac{\pi_t^F(L, s)}{\pi_t^F(H, g)} \to K(H, g) - K(L, s) = \frac{1}{2} \log \frac{25}{24} > 0 \qquad \mathbb{P}^*\text{-a.s.} \tag{6}$$

The full Bayesian posterior therefore favors the true diagonal state at an asymptotically positive rate, and its marginal report satisfies $\mathsf{M}(\pi_t^F) \to (\delta_L, \delta_s)$.

Recursive marginal learning does not preserve the diagonal comparison in (6). Each coordinate is updated using the current marginal assessment of the other coordinate. Near (H, g), changing the project type alone appears unfavorable, and changing the diagnostic regime alone also appears unfavorable. Correcting the false interpretation requires revising both coordinates together, but the marginal record does not retain the posterior association needed to carry that joint comparison forward.

The example reveals the central geometry of the paper. Asymptotically, full Bayesian posterior odds favor joint states with lower KL divergence from the true signal distribution. Recursive marginal learning instead generates coordinate-wise comparisons because each coordinate is evaluated using the recorded assessment of the other. A false state can consequently become self-reinforcing even though another state fits the evidence better globally. The next section formalizes this criterion and characterizes when a pure state can be selected.

4 Coordinate-wise Selection

We now formalize the coordinate-wise comparisons illustrated by the binary example. The recursive log-odds dynamics show that the evidence used to evaluate either coordinate depends on the learner's current marginal assessment of the other. We use this observation to characterize convergence to pure records. Such convergence can occur with positive probability only when the associated state fits the evidence at least as well as every alternative obtained by changing one coordinate. Conversely, if all these comparisons are strict, records that sufficiently favor both components of the state converge to its pure record with high probability.

Throughout this section, let $q_t = (\alpha_t, \beta_t)$ denote the recursive marginal record generated from $q_0 \in Q$ through (3). All probabilities are taken under the true signal law $\mathbb{P}^*$.

4.1 Effective likelihoods and coordinate-wise fit

Given a record $q = (\alpha, \beta) \in \overline{Q}$, define the effective likelihoods

$$\widetilde{P}_A(y \mid a; \beta) := \sum_{b \in B} \beta(b) P_{ab}(y), \qquad \widetilde{P}_B(y \mid b; \alpha) := \sum_{a \in A} \alpha(a) P_{ab}(y).$$

The effective likelihood $\widetilde{P}_A(\cdot \mid a; \beta)$ evaluates a by averaging over the current marginal assessment of B, and $\widetilde{P}_B(\cdot \mid b; \alpha)$ is defined symmetrically.

For every distinct $a, a' \in A$ and $b, b' \in B$, from the update rule (3), the recursive update satisfies

$$\begin{aligned} \log \frac{\alpha_{t+1}(a)}{\alpha_{t+1}(a')} &= \log \frac{\alpha_t(a)}{\alpha_t(a')} + \log \frac{\widetilde{P}_A(Y_{t+1} \mid a; \beta_t)}{\widetilde{P}_A(Y_{t+1} \mid a'; \beta_t)}, \\ \log \frac{\beta_{t+1}(b)}{\beta_{t+1}(b')} &= \log \frac{\beta_t(b)}{\beta_t(b')} + \log \frac{\widetilde{P}_B(Y_{t+1} \mid b; \alpha_t)}{\widetilde{P}_B(Y_{t+1} \mid b'; \alpha_t)}. \end{aligned} \tag{7}$$

Thus the evidence used to compare two values of the first coordinate depends on the current record over the second coordinate, and the evidence used to compare two values of the second coordinate depends on the current record over the first. Each marginal assessment therefore determines the likelihood comparison used to update the other.

Fix a candidate state $\bar{x} = (\bar{a}, \bar{b})$. For every $a \neq \bar{a}$ and $b \neq \bar{b}$, define the expected marginal log-odds changes

$$\begin{aligned} \Phi_a(\beta) &:= \sum_{y \in Y} P^*(y) \log \frac{\widetilde{P}_A(y \mid \bar{a}; \beta)}{\widetilde{P}_A(y \mid a; \beta)}, \\ \Psi_b(\alpha) &:= \sum_{y \in Y} P^*(y) \log \frac{\widetilde{P}_B(y \mid \bar{b}; \alpha)}{\widetilde{P}_B(y \mid b; \alpha)}. \end{aligned}$$

These quantities are the expected one-period changes in the corresponding marginal log odds under the true signal distribution P^*.

At $\beta = \delta_{\bar{b}}$ and $\alpha = \delta_{\bar{a}}$, the effective likelihoods reduce to the state-dependent likelihoods P_{ab}. Hence[5]

$$\Phi_a(\delta_{\bar{b}}) = K(a, \bar{b}) - K(\bar{a}, \bar{b}), \qquad \Psi_b(\delta_{\bar{a}}) = K(\bar{a}, b) - K(\bar{a}, \bar{b}). \tag{8}$$

Equation (8) identifies the comparisons that govern learning when the record favors $\bar{x}$. As

[5] For example,

$$\Phi_a(\delta_{\bar{b}}) = \sum_{y \in Y} P^*(y) \log \frac{P_{\bar{a}\bar{b}}(y)}{P_{a\bar{b}}(y)} = D_{\mathrm{KL}}(P^* \| P_{a\bar{b}}) - D_{\mathrm{KL}}(P^* \| P_{\bar{a}\bar{b}}) = K(a, \bar{b}) - K(\bar{a}, \bar{b}).$$

The identity for Ψ_b is analogous.

the assessment of B approaches $\delta_{\bar{b}}$, the update of A increasingly compares $(\bar{a}, \bar{b})$ with states $(a, \bar{b})$. As the assessment of A approaches $\delta_{\bar{a}}$, the update of B increasingly compares $(\bar{a}, \bar{b})$ with states $(\bar{a}, b)$. States that differ from $\bar{x}$ in both coordinates do not enter either comparison directly.

Definition 1. A state $\bar{x} = (\bar{a}, \bar{b})$ is a *weak coordinate-wise KL minimum* if

$$K(\bar{a}, \bar{b}) \leq K(a, \bar{b}) \quad \text{for every } a \neq \bar{a}, \qquad K(\bar{a}, \bar{b}) \leq K(\bar{a}, b) \quad \text{for every } b \neq \bar{b}. \tag{9}$$

It is a *strict coordinate-wise KL minimum* if all the inequalities in (9) are strict.

Coordinate-wise KL minimality is weaker than global KL minimality.[6] A state may fit the evidence better than every alternative obtained by changing one coordinate even though another state, differing in both coordinates, fits the evidence better overall. A full Bayesian learner retains the direct posterior comparison between these joint states. Under recursive marginal learning, each coordinate is instead evaluated using the current marginal assessment of the other.

Under identification, the true state $x^* = (a^*, b^*)$ is a strict coordinate-wise KL minimum because $K(a^*, b^*) = 0$ while every false state has strictly positive KL divergence.[7]

4.2 State selection and reinforcement

We say that a state $x = (a, b)$ is *selected from* q_0 if $\mathbb{P}^* (q_t \to (\delta_a, \delta_b)) > 0$. If $x \neq x^*$, its selection constitutes *false certainty.*

For $C > 0$ and a candidate state $x = (a, b)$, define

$$\mathcal{G}_C(x) := \left\{ (\alpha, \beta) \in Q : \min_{a' \neq a} \log \frac{\alpha(a)}{\alpha(a')} \geq C, \ \min_{b' \neq b} \log \frac{\beta(b)}{\beta(b')} \geq C \right\}.$$

A record in $\mathcal{G}_C(x)$ favors both coordinates of x by at least C in log odds. In particular, $q \in \mathcal{G}_C(x)$ implies

$$\alpha(a) \geq \frac{1}{1 + (|A| - 1)e^{-C}}, \qquad \beta(b) \geq \frac{1}{1 + (|B| - 1)e^{-C}}.$$

Larger values of C therefore describe records that place greater probability on both components of the same state.

[6] We define the global KL minimality $\bar{x} = (\bar{a}, \bar{b})$ if $K(\bar{a}, \bar{b}) \leq K(a, b)$ for any $(a, b) \in X$.

[7] Under identification, any false weak coordinate-wise KL minimum must differ from the true state in both coordinates. If it shared either coordinate with the truth, the true state would be one of its one-coordinate alternatives and would fit the evidence strictly better.

The next theorem links coordinate-wise fit to state selection and reinforcement.

Theorem 1. *Fix $\bar{x} = (\bar{a}, \bar{b})$.*

(i) Necessity. *For any $q_0 \in Q$, if $\mathbb{P}^* (q_t \to (\delta_{\bar{a}}, \delta_{\bar{b}})) > 0$, then $\bar{x}$ is a weak coordinate-wise KL minimum.*

(ii) Reinforcement. *If $\bar{x}$ is a strict coordinate-wise KL minimum, then there exist $C, \gamma > 0$ such that, with $d := |A| + |B| - 2$, for every $r > 0$ and every $q_0 \in \mathcal{G}_{C+r}(\bar{x})$,*

$$\mathbb{P}^* (q_t \to (\delta_{\bar{a}}, \delta_{\bar{b}})) \geq 1 - de^{-\gamma r}.$$

Part (i) restricts the states that can be selected. If the evidence favors changing either coordinate while holding the other fixed, convergence to the corresponding pure record has probability zero. Every selected state must therefore fit the evidence at least as well as each of its one-coordinate alternatives. This condition does not require the state to fit the evidence better than every joint alternative, because a better-fitting state may differ in both coordinates and therefore may not enter either marginal comparison directly.

Part (ii) shows that a strict coordinate-wise KL minimum becomes strongly reinforcing once both of its components receive enough support. The constant C identifies a region in which all relevant expected marginal log-odds changes favor $\bar{x}$, while r measures the additional initial support beyond that threshold. If $q_0 \in \mathcal{G}_{C+r}(\bar{x})$, the probability of convergence to the pure record associated with $\bar{x}$ is at least $1 - de^{-\gamma r}$, which approaches one as r increases.

The reinforcement is generated jointly by the two marginal assessments. Support for $\bar{b}$ makes the update of the first coordinate favor $\bar{a}$, while support for $\bar{a}$ makes the update of the second coordinate favor $\bar{b}$. Each recorded assessment can therefore sustain the likelihood comparison that reinforces the other.

When there are no one-coordinate KL ties, the theorem yields an exact characterization of pure-state selection.

Corollary 1. *Fix $\bar{x} = (\bar{a}, \bar{b})$, and suppose*

$$K(\bar{a}, \bar{b}) \neq K(a, \bar{b}) \quad \text{for every } a \neq \bar{a}, \qquad K(\bar{a}, \bar{b}) \neq K(\bar{a}, b) \quad \text{for every } b \neq \bar{b}.$$

Then the following statements are equivalent.

(i) *The state $\bar{x}$ is a strict coordinate-wise KL minimum.*

(ii) *For every $\varepsilon \in (0, 1)$, there exists $C_\varepsilon > 0$ such that every $q_0 \in \mathcal{G}_{C_\varepsilon}(\bar{x})$ satisfies*

$$\mathbb{P}^* (q_t \to (\delta_{\bar{a}}, \delta_{\bar{b}})) \geq 1 - \varepsilon.$$

(iii) *There exists a full-support initial record* $q_0 \in Q$ *such that* $\mathbb{P}^* \left(q_t \to (\delta_{\bar{a}}, \delta_{\bar{b}})\right) > 0$.

Away from one-coordinate ties, a state can be selected from some full-support initial record if and only if it fits the evidence strictly better than every alternative obtained by changing only its first coordinate or only its second. Records that favor both components strongly enough then converge to its pure record with arbitrarily high probability. If the state loses any one-coordinate comparison, it cannot be selected. A false state may therefore survive every isolated revision even though another state, requiring both coordinates to change, fits the evidence better globally.

5 Global Selection and Robust False Certainty

The reinforcement of Theorem 1 requires a sufficiently favorable region of the prior. We now move from this local reinforcement to learning from arbitrary full-support initial records. Fix a state $x = (a, b)$. The relevant question is whether an interpretation that continues to receive support in both marginal assessments can be moved, by a finite sequence of observations, into a region where the reinforcement result of the previous section applies. We formulate a uniform accessibility condition that requires this possibility whenever the product mass $\alpha(a)\beta(b)$ is bounded away from zero.

We first combine accessibility with reinforcement to establish a persistence result. A strict coordinate-wise KL minimum that is finitely accessible either becomes the learner's limiting state or eventually loses support in at least one marginal. We then apply this result to an open neighborhood of the binary example. Throughout that neighborhood, the recursive marginal learner converges almost surely to either the true or the false diagonal state from every full-support initial record, and each limit occurs with positive probability.

Let

$$\mathcal{P} := \prod_{(a,b)\in X} \Delta(Y)$$

denote the space of full-support likelihood systems, equipped with the relative Euclidean topology inherited from $\prod_{(a,b)\in X} \mathbb{R}^{|Y|}$. Fix the true state $x^* = (a^*, b^*)$. For $P = (P_{ab})_{(a,b)\in X} \in \mathcal{P}$, write

$$K^P(a, b) := D_{\mathrm{KL}}(P_{a^*b^*} \| P_{ab}),$$

and let U_y^P denote the recursive update map induced by P. Weak and strict coordinate-wise KL minima under P are defined as in Definition 1, with K^P replacing K. Let $\mathbb{P}^P$ and $\mathbb{E}^P$ denote probability and expectation when signals are independently generated according to $P_{a^*b^*}$, and let $\mathbb{P}_q^P$ emphasize that the recursive process begins from the deterministic record

q.

5.1 Accessibility and persistence

For a state $x = (a, b)$ and a record $q = (\alpha, \beta) \in \overline{Q}$, define the probability assigned to x by the product reconstruction:

$$m_x(q) := R(q)(a, b) = \alpha(a)\beta(b).$$

The product mass $m_x(q)$ records whether both components of x continue to receive positive probability in the corresponding marginal assessments.[8]

For a finite signal word $w = (y_1, \ldots, y_T)$, write

$$U_w^P := U_{y_T}^P \circ \cdots \circ U_{y_1}^P,$$

with the empty word inducing the identity map.

Definition 2. Fix $P \in \mathcal{P}$. A state $x \in X$ is *finitely accessible away from zero product mass* if, for every $\varepsilon \in (0, 1)$ and every $C > 0$, there exists a finite signal word w such that

$$U_w^P(q) \in \mathcal{G}_C(x)$$

for every $q \in Q$ satisfying $m_x(q) \geq \varepsilon$.

Finite accessibility requires a single finite signal sequence that works uniformly across all records assigning at least ε product mass to x. For any target level C, the same sequence moves every such record into $\mathcal{G}_C(x)$. The state need not initially be favored in either marginal assessment; it is enough that both of its components retain sufficiently positive probability. The condition therefore captures whether continued support for an interpretation leaves open a finite path by which the two marginal assessments can become jointly concentrated around it.

If x is also a strict coordinate-wise KL minimum, accessibility can move the record into a region where its two components reinforce one another. Theorem 1 then gives a positive lower bound on the probability of convergence to the associated pure record.

Theorem 2. *Fix $P \in \mathcal{P}$, and let $x = (a, b)$ be a strict coordinate-wise KL minimum under P that is finitely accessible away from zero product mass. Let $E_x := \{q_t \to (\delta_a, \delta_b)\}$. For*

[8] It is bounded away from zero only if neither component has been nearly discarded.

every $\varepsilon \in (0,1)$*, there exists* $c_{x,\varepsilon} > 0$ *such that*

$$m_x(q) \geq \varepsilon \quad \Longrightarrow \quad \mathbb{P}^P_q(E_x) \geq c_{x,\varepsilon}$$

for every $q \in Q$*. Consequently, for every* $q_0 \in Q$*, (i)* $\mathbb{P}^P_{q_0}(E_x) > 0$*, and (ii)* $m_x(q_t) \to 0$*,* $\mathbb{P}^P_{q_0}$*-a.s. on* E^c_x*.*

The first conclusion extends the local reinforcement result to records that need not initially favor x. If x retains product mass at least ε, finite accessibility provides a signal sequence that can move the record into a reinforcing region, so the probability of eventual selection of x is bounded below by $c_{x,\varepsilon} > 0$. Importantly, this lower bound is uniform over all records satisfying $m_x(q) \geq \varepsilon$. Thus, whenever x continues to receive nonnegligible support in both marginal assessments, it continues to have a probability bounded away from zero of eventually being selected.

The second conclusion describes what must happen on paths where x is not selected. On E^c_x, $\alpha_t(a)\beta_t(b) \to 0$ almost surely. Thus, if the learner does not converge to (δ_a, δ_b), the two components of x cannot both continue to receive probabilities bounded away from zero. At least one of $\alpha_t(a)$ or $\beta_t(b)$ must become arbitrarily small. Intuitively, if both components continued to receive nonnegligible support infinitely often, then at each such date the process would retain a probability bounded away from zero of eventually selecting x, which is inconsistent with being on a path where x is never selected.

5.2 A robust false-certainty environment

Return to the binary environment of Section 3. Let $P^0 = (P^0_{ab})_{(a,b)\in X} \in \mathcal{P}$ denote the baseline likelihood system

$$P^0_{Hg}(1) = \frac{3}{5}, \qquad P^0_{Hs}(1) = \frac{1}{10}, \qquad P^0_{Lg}(1) = \frac{9}{10}, \qquad P^0_{Ls}(1) = \frac{1}{2}.$$

For every P near P^0, retain $x^* = (L, s)$ as the true state and let signals be independently generated according to P_{Ls}.

Fix $q_0 \in Q$. Initialize the full Bayesian learner at $\pi^F_0 = R(q_0)$ and the recursive marginal learner at q_0.

For $\zeta > 0$, define

$$Q_\zeta := \left\{ (\alpha, \beta) \in Q : \left| \log \frac{\alpha(H)}{\alpha(L)} \right| \leq \zeta, \ \left| \log \frac{\beta(g)}{\beta(s)} \right| \leq \zeta \right\}.$$

The set Q_ζ contains the initial records whose two marginal log odds are bounded in magnitude by ζ. Every full-support record belongs to Q_ζ for some finite ζ.

Theorem 3. *There exists an open neighborhood $\mathcal{O}$ of P^0 such that the following statements hold.*

(i) *For every $P \in \mathcal{O}$ and every $q_0 \in Q$, the full Bayesian learner learns the true state:*

$$\pi_t^F \to \delta_{(L,s)} \qquad \mathbb{P}^P\text{-a.s.}$$

(ii) *For every $P \in \mathcal{O}$ and every $q_0 \in Q$, the recursive marginal record converges almost surely to one of the two diagonal records:*

$$\mathbb{P}_{q_0}^P \left(q_t \to (\delta_L, \delta_s) \text{ or } q_t \to (\delta_H, \delta_g) \right) = 1.$$

Both limiting events have strictly positive $\mathbb{P}_{q_0}^P$-probability. Moreover, for every $\zeta > 0$, there exists $\eta_\zeta > 0$ such that

$$\inf_{P \in \mathcal{O}} \inf_{q_0 \in Q_\zeta} \mathbb{P}_{q_0}^P \left(q_t \to (\delta_H, \delta_g) \right) \geq \eta_\zeta.$$

Part (i) shows that identification is preserved throughout the neighborhood. A full Bayesian learner who begins from any full-support prior learns the true state (L, s). Her marginal report therefore satisfies

$$\mathsf{M}(\pi_t^F) \to (\delta_L, \delta_s) \qquad \mathbb{P}^P\text{-a.s.}$$

Reporting the marginals is harmless when those marginals continue to be computed from the complete joint posterior and the joint posterior remains available for future updating.

Part (ii) shows that recursively reusing the same marginal assessments changes the long-run outcome. From every full-support initial record, the recursive marginal learner eventually becomes certain of either the true state (L, s) or the false state (H, g), and each outcome has positive probability. The false outcome is not confined to records initially close to (H, g). Full support leaves both components of the false state available, and a suitable finite sequence of observations can move the record into a region where they reinforce one another.

The uniform bound over Q_ζ strengthens this conclusion. For any fixed bound on the initial marginal log odds, the probability of false certainty is bounded away from zero uniformly over the initial record and the likelihood system within $\mathcal{O}$. As ζ increases, Q_ζ includes records that assign increasingly small probability to H or g, so the lower bound η_ζ may decrease.

The divergence occurs along the same realized signal histories. For every fixed $P \in \mathcal{O}$

and $q_0 \in Q$, part (i) holds almost surely while false convergence in part (ii) has positive probability. Hence there is a positive-probability set of histories on which

$$\pi_t^F \to \delta_{(L,s)}, \qquad \mathsf{M}(\pi_t^F) \to (\delta_L, \delta_s), \qquad q_t \to (\delta_H, \delta_g).$$

On these histories, the two learners begin from the same product belief, observe the same signals, know the same likelihood system, and apply Bayes' rule within each period. Their conclusions differ only because the full Bayesian learner carries posterior association forward, while the recursive marginal learner replaces each within-period posterior with the product of its marginals before the next signal is processed.

The theorem therefore identifies an intertemporal cost of record design. A marginal record can exactly report the within-period posterior assessment of each coordinate while omitting the association needed to combine those assessments with future evidence. The next section identifies the likelihood systems under which no signal creates such omitted information.

6 The Exact Lossless Boundary

The preceding sections show that recursive marginal learning can redirect inference because Bayesian updating creates posterior association that is not retained for future updating. This section identifies exactly when no relevant information is lost. Recursive marginal learning reproduces full Bayesian learning from every product prior if and only if, for each signal realization, its likelihood can be written as the product of two positive terms: one depending only on the first state coordinate and the other depending only on the second.

Under this condition, updating from a product belief never creates posterior association between the state coordinates. The full posterior therefore remains determined by its two marginals, and recursive marginal learning reproduces full Bayesian learning after every finite signal history. When both state coordinates are binary and the condition fails, the omitted information has one dimension. The two posterior marginals together with one additional association coordinate then recover the entire joint posterior.

6.1 No coordinate interaction

We say that the likelihood system has *no coordinate interaction* if, for every signal realization $y \in Y$, there exist functions $f_A^y : A \to (0,\infty)$ and $f_B^y : B \to (0,\infty)$ such that

$$P_{ab}(y) = f_A^y(a) f_B^y(b) \qquad \text{for every } a \in A,\ b \in B. \tag{10}$$

Thus, for a fixed signal realization, its likelihood separates into one positive term depending only on the first coordinate and another depending only on the second.

Condition (10) implies that observing y does not create posterior association when the prior is a product belief. The likelihood contribution changes the relative probabilities assigned to values of A and B separately, so Bayesian updating preserves the product form.

Theorem 4. *The following statements are equivalent.*

(i) *The likelihood system has no coordinate interaction.*

(ii) *For every $q \in Q$ and every $y \in Y$, the posterior $\mathsf{B}_y(R(q))$ is a product belief.*

(iii) *For every $q_0 \in Q$ and every finite signal history, if the full Bayesian learner is initialized at $\pi_0^F = R(q_0)$, then $R(q_t) = \pi_t^F$ at every date along that history.*

Moreover, under these equivalent conditions, every weak coordinate-wise KL minimum is a global KL minimizer. Under identification, the true state is the unique weak coordinate-wise KL minimum.

The theorem gives the exact condition under which the two marginals contain everything needed for future Bayesian updating from a product prior. Under the no coordinate interaction, every posterior generated from a product belief remains a product belief. Marginalization therefore discards no information, and the product reconstructed from the two posterior marginals is exactly the full Bayesian posterior.

If the condition fails, some signal realization creates posterior association from every full-support product belief. The resulting marginals still exactly describe the within-period posterior assessment of each coordinate separately, but they no longer determine the joint posterior from which they were obtained. Replacing that posterior with the product of its marginals can therefore change how later signals are interpreted. The two procedures may still agree after particular signal histories, but they no longer agree after every finite history from every product prior.

The same condition connects coordinate-wise and global fit. Under the condition (10), the KL divergence has the additive form

$$K(a, b) = c + \kappa_A(a) + \kappa_B(b),$$

where

$$c := \sum_{y \in Y} P^*(y) \log P^*(y),$$
$$\kappa_A(a) := -\sum_{y \in Y} P^*(y) \log f_A^y(a),$$
$$\kappa_B(b) := -\sum_{y \in Y} P^*(y) \log f_B^y(b).$$

A state $(\bar{a}, \bar{b})$ is therefore a weak coordinate-wise KL minimum if and only if $\bar{a}$ minimizes κ_A and $\bar{b}$ minimizes κ_B. Such a state also minimizes $K(a,b)$ over all joint states. Under identification, no false state can satisfy the weak coordinate-wise criterion.

Starting from a product belief, marginal records therefore preserve everything needed for future Bayesian updating precisely when no signal realization creates posterior association between the state coordinates. When the likelihood contribution of every signal separates as in the condition above, an organization can maintain the two assessments separately without losing information needed for later inference.

6.2 Exact binary augmentation

We now specialize to a 2×2 state space. The result applies to every full-support likelihood system on such a state space, not only to the example in Section 3. Let

$$A = \{a_0, a_1\}, \qquad B = \{b_0, b_1\}.$$

A joint belief over four states has three degrees of freedom, whereas its two binary marginals contain only two.[9]

For any $\pi \in \Delta(X)$, define its posterior association coordinate by

$$\ell(\pi) := \log \frac{\pi(a_1, b_1)\pi(a_0, b_0)}{\pi(a_1, b_0)\pi(a_0, b_1)}. \tag{11}$$

The log cross-ratio in (11) compares the relative posterior weight assigned to the matched pairs (a_1, b_1) and (a_0, b_0) with the weight assigned to the crossed pairs (a_1, b_0) and (a_0, b_1).

[9] Write the four joint probabilities as $\pi_{00}, \pi_{01}, \pi_{10}, \pi_{11}$. They satisfy $\pi_{00} + \pi_{01} + \pi_{10} + \pi_{11} = 1$, so only three of the four probabilities are independent. By contrast, each binary marginal is determined by one number, for example

$$\pi_A(a_1) = \pi_{10} + \pi_{11}, \qquad \pi_B(b_1) = \pi_{01} + \pi_{11},$$

so the two marginals provide only two independent numbers. One additional scalar is therefore needed to determine the joint belief.

Equivalently,

$$\exp\bigl(\ell(\pi)\bigr) = \frac{\pi(a_1 \mid b_1)\pi(a_0 \mid b_0)}{\pi(a_1 \mid b_0)\pi(a_0 \mid b_1)}.$$

Thus $\ell(\pi) = 0$ if and only if the two coordinates are independent. A positive value assigns relatively greater weight to the matched pairs, while a negative value assigns relatively greater weight to the crossed pairs.

For each signal $y \in Y$, define its likelihood interaction by

$$\lambda_y := \log \frac{P_{a_1 b_1}(y) P_{a_0 b_0}(y)}{P_{a_1 b_0}(y) P_{a_0 b_1}(y)}. \tag{12}$$

Corollary 2. *In a full-support 2×2 environment, the pair $\bigl(\mathsf{M}(\pi), \ell(\pi)\bigr)$ uniquely determines every joint belief $\pi \in \Delta(X)$. Moreover, for every $y \in Y$,*

$$\ell\bigl(\mathsf{B}_y(\pi)\bigr) = \ell(\pi) + \lambda_y, \qquad \ell\bigl(R(\mathsf{M}(\pi))\bigr) = 0. \tag{13}$$

Consequently, if the full Bayesian learner begins from a product prior $\pi_0^F = R(q_0)$, then

$$\ell(\pi_t^F) = \sum_{\tau=1}^{t} \lambda_{Y_\tau}, \tag{14}$$

and the pair $\bigl(\mathsf{M}(\pi_t^F), \ell(\pi_t^F)\bigr)$ uniquely determines the full Bayesian posterior at every date. Finally, the marginal record alone reproduces full Bayesian learning from every product prior if and only if $\lambda_y = 0$ for every $y \in Y$, equivalently, if and only if the likelihood system has no coordinate interaction.

The corollary gives an exact completion of the binary marginal record. The two marginals describe the current assessment of each coordinate, while $\ell(\pi)$ records their posterior association. Together, these quantities uniquely determine the joint posterior. A learner who retains them can reconstruct the full posterior before the next signal arrives, apply Bayes' rule, and then store the updated marginals together with the updated association coordinate. No additional information from the earlier signal history is required.

Full Bayesian learning carries posterior association forward. By the equations in (13), observing signal y adds λ_y to the existing log cross-ratio $\ell(\pi)$. Starting from a product prior, the effects of successive signals therefore accumulate according to (14). Recursive marginal learning instead replaces the posterior with the product of its marginals after every update and therefore resets the association coordinate to zero before the next signal is processed.

From the definition of (12), the condition $\lambda_y = 0$ is equivalent to

$$P_{a_1 b_1}(y) P_{a_0 b_0}(y) = P_{a_1 b_0}(y) P_{a_0 b_1}(y).$$

Thus the marginal record alone reproduces full Bayesian learning from every product prior exactly when this equality holds for every $y \in Y$. If it fails for some signal realization, that signal creates posterior association that cannot in general be recovered from the two marginals alone.

7 Conclusion

This paper studies learning through recursively reused marginal records. The learner observes every signal, knows the correct likelihood system, and applies Bayes' rule to the reconstructed product belief at each date, but carries forward only the two resulting posterior marginals. The record therefore reports its two within-period marginal assessments exactly while potentially omitting information needed for future inference. We first show that because each coordinate is evaluated using the current marginal assessment of the other coordinate, every selected state must be a weak coordinate-wise KL minimum. Conversely, every strict coordinate-wise KL minimum attracts records that sufficiently favor both of its components, with the probability of convergence approaching one as that initial support increases.

In the binary environment studied here, this selection mechanism governs the entire long-run process throughout an open neighborhood of the numerical example. From every full-support initial record, the recursive marginal learner converges almost surely to either the true diagonal state or the false diagonal state, and each limit occurs with positive probability. The probability of false certainty is uniformly bounded away from zero on every bounded log-odds class, even though the full Bayesian learner converges to the truth from the same product prior and along the same realized signal histories.

Recursive marginal learning reproduces full Bayesian learning from every product prior exactly when, for each signal realization, its likelihood factors into one positive term depending only on the first state coordinate and another depending only on the second. Under this no-coordinate-interaction condition, updating from a product belief never creates posterior association, so the full posterior remains determined by its marginals. When the condition fails, the two marginals can still report each coordinate exactly while omitting their posterior association, and this omitted information can affect how later signals are interpreted. With two binary coordinates, retaining one additional association coordinate together with the two marginals is sufficient to reconstruct the full posterior exactly. The broader implication is that an informational record should be evaluated not only by what it reports when it is produced, but also by whether it preserves the relationships needed to interpret future evidence.

A Proofs

Throughout Appendix A, recall that $\Delta(S)$ and Q contain full-support distributions, while $\overline{\Delta}(S)$ and $\overline{Q}$ include boundary distributions. Let $\mathcal{F}_t := \sigma(Y_1, \dots, Y_t)$ denote the natural filtration. Unless a likelihood system is displayed explicitly, probabilities, expectations, and almost-sure statements are taken under $\mathbb{P}^*$. In abstract probability lemmas, $\mathbb{P}$ and $\mathbb{E}$ denote probability and expectation on the underlying filtered probability space.

When the likelihood system varies, recall that $\mathcal{G}_C(x)$ is defined in Section 4, while $\mathcal{P}$, K^P, U_y^P, $\mathbb{P}^P$, $\mathbb{E}^P$, and $m_x(q)$ are defined in Section 5. For $q = (\alpha, \beta) \in \overline{Q}$, define the effective likelihoods

$$\widetilde{P}_A^P(y \mid a; \beta) := \sum_{b \in B} \beta(b) P_{ab}(y), \qquad \widetilde{P}_B^P(y \mid b; \alpha) := \sum_{a \in A} \alpha(a) P_{ab}(y).$$

Whenever P is fixed, its superscript is omitted. The notation $\mathbb{P}_q^P$ is used when a recursively generated event is viewed as a function of the deterministic initial record q; the underlying signal law remains $\mathbb{P}^P$. For a subset S of a finite-dimensional simplex or likelihood space, $\overline{S}$ denotes its closure in the corresponding relative Euclidean topology. For a nonempty set $S \subseteq \overline{Q}$, write

$$\operatorname{dist}(q, S) := \inf_{z \in S} \|q - z\|,$$

where $\|\cdot\|$ is any norm.

A.1 Proofs of the main-text results

Proof of Proposition 1

Fix $\pi \in \Delta(X)$ and $q = (\alpha, \beta) \in \overline{Q}$. If $\alpha(a) = 0$ for some a or $\beta(b) = 0$ for some b, then $R(q)$ assigns zero probability to a state to which π assigns positive probability, so $D_{\mathrm{KL}}(\pi \| R(q)) = +\infty$. It therefore suffices to consider $q \in Q$. For such q,

$$D_{\mathrm{KL}}\big(\pi \| R(q)\big) = D_{\mathrm{KL}}\big(\pi \| R(\mathsf{M}(\pi))\big) + D_{\mathrm{KL}}(\pi_A \| \alpha) + D_{\mathrm{KL}}(\pi_B \| \beta). \tag{15}$$

Indeed,

$$\log \frac{\pi(a,b)}{\alpha(a)\beta(b)} = \log \frac{\pi(a,b)}{\pi_A(a)\pi_B(b)} + \log \frac{\pi_A(a)}{\alpha(a)} + \log \frac{\pi_B(b)}{\beta(b)}.$$

Summing against $\pi(a,b)$ gives (15). The first term on the right-hand side is independent of q, while the last two terms are nonnegative and vanish simultaneously if and only if $\alpha = \pi_A$ and $\beta = \pi_B$. Hence $\mathsf{M}(\pi)$ is the unique minimizer. ■

Proof of Proposition 2

Fix $x = (a,b) \neq x^*$, and write $P_x := P_{ab}$. By (2),

$$\frac{1}{t} \log \frac{\pi_t^F(x)}{\pi_t^F(x^*)} = \frac{1}{t} \log \frac{\pi_0^F(x)}{\pi_0^F(x^*)} + \frac{1}{t} \sum_{\tau=1}^{t} \log \frac{P_x(Y_\tau)}{P^*(Y_\tau)}.$$

The summands are bounded because Y is finite and both likelihoods have full support. The strong law of large numbers therefore gives[10]

$$\frac{1}{t} \sum_{\tau=1}^{t} \log \frac{P_x(Y_\tau)}{P^*(Y_\tau)} \to \sum_{y \in Y} P^*(y) \log \frac{P_x(y)}{P^*(y)} = -K(x) \qquad \mathbb{P}^*\text{-a.s.}$$

Since the initial term converges to zero,

$$\frac{1}{t} \log \frac{\pi_t^F(x)}{\pi_t^F(x^*)} \to -K(x) < 0 \qquad \mathbb{P}^*\text{-a.s.}$$

Identification gives $K(x) > 0$ for every $x \neq x^*$, so $\pi_t^F(x)/\pi_t^F(x^*) \to 0$ almost surely for every false state. Since X is finite,

$$\pi_t^F(x^*) = \frac{1}{1 + \sum_{x \neq x^*} \pi_t^F(x)/\pi_t^F(x^*)} \to 1 \qquad \mathbb{P}^*\text{-a.s.}$$

Thus $\pi_t^F \to \delta_{(a^*,b^*)}$, and continuity of marginalization gives $\mathsf{M}(\pi_t^F) \to (\delta_{a^*}, \delta_{b^*})$ almost surely. ■

Proof of Theorem 1

Let $P := (P_{ab})_{(a,b) \in X} \in \mathcal{P}$ denote the fixed likelihood system of the model. We first state one auxiliary result.

[10] The strong law of large numbers states that if $(X_t)_{t \geq 1}$ are independent and identically distributed random variables with $\mathbb{E}[|X_1|] < \infty$, then $\frac{1}{t} \sum_{s=1}^{t} X_s \to \mathbb{E}[X_1]$ almost surely.

Lemma A.1. *Let $(\xi_t)_{t\geq 1}$ be martingale differences with respect to $(\mathcal{F}_t)$. If $|\xi_t| \leq L$ almost surely for every t, then $t^{-1}\sum_{s=1}^{t}\xi_s \to 0$ almost surely.*

For part (i), suppose weak coordinate-wise KL minimality fails along the A-coordinate, so there exists $a \neq \bar{a}$ such that $K(\bar{a},\bar{b}) > K(a,\bar{b})$. Recall

$$\Phi_a(\beta) := \sum_{y\in Y} P^*(y)\log\frac{\widetilde{P}_A(y \mid \bar{a};\beta)}{\widetilde{P}_A(y \mid a;\beta)}.$$

At $\beta = \delta_{\bar{b}}$,

$$\Phi_a(\delta_{\bar{b}}) = \sum_{y\in Y} P^*(y)\log\frac{P_{\bar{a}\bar{b}}(y)}{P_{a\bar{b}}(y)} = K(a,\bar{b}) - K(\bar{a},\bar{b}) < 0.$$

Continuity of Φ_a on $\overline{\Delta}(B)$ gives a neighborhood N of $\delta_{\bar{b}}$ and $\kappa > 0$ such that $\Phi_a(\beta) \leq -\kappa$ for every $\beta \in N$.

Fix $q_0 \in Q$, and define

$$E := \{q_t \to (\delta_{\bar{a}}, \delta_{\bar{b}})\}, \qquad Z_t^a := \log\frac{\alpha_t(\bar{a})}{\alpha_t(a)}.$$

On E, there is a finite random date T such that $\beta_t \in N$, and hence $\Phi_a(\beta_t) \leq -\kappa$, for every $t \geq T$. At the same time,

$$Z_t^a \to +\infty \tag{16}$$

on E.

The first identity in (7) gives

$$Z_{t+1}^a - Z_t^a = \log\frac{\widetilde{P}_A(Y_{t+1} \mid \bar{a};\beta_t)}{\widetilde{P}_A(Y_{t+1} \mid a;\beta_t)}, \qquad \mathbb{E}^*[Z_{t+1}^a - Z_t^a \mid \mathcal{F}_t] = \Phi_a(\beta_t).$$

Define $\xi_{t+1}^a := Z_{t+1}^a - Z_t^a - \Phi_a(\beta_t)$ and $M_t^a := \sum_{s=0}^{t-1}\xi_{s+1}^a$. Then

$$Z_t^a = Z_0^a + \sum_{s=0}^{t-1}\Phi_a(\beta_s) + M_t^a. \tag{17}$$

Because all state-dependent likelihoods P_{ab} have full support and A, B, and Y are finite, all effective log-likelihood ratios and hence the martingale differences ξ_{t+1}^a are uniformly bounded. Lemma A.1 therefore implies that $\Omega_0 := \{M_t^a/t \to 0\}$ has probability one.

On $E \cap \Omega_0$, the finite initial segment before T is asymptotically negligible, so

$$\limsup_{t\to\infty} \frac{1}{t} \sum_{s=0}^{t-1} \Phi_a(\beta_s) \leq -\kappa.$$

Dividing (17) by t gives $\limsup_{t\to\infty} Z_t^a/t \leq -\kappa$, contradicting (16). Hence $\mathbb{P}^*(E) = 0$. The argument for the B-coordinate is symmetric, proving part (i).

For part (ii), we use the following uniform local-attraction result.

Proposition A.1. *Let $\mathcal{K} \subset \mathcal{P}$ be compact and fix $\bar{x} = (\bar{a}, \bar{b})$. Suppose*

$$\kappa_0 := \inf_{P\in\mathcal{K}} \min \left\{ \min_{a\neq\bar{a}} [K^P(a, \bar{b}) - K^P(\bar{a}, \bar{b})],\ \min_{b\neq\bar{b}} [K^P(\bar{a}, b) - K^P(\bar{a}, \bar{b})] \right\} > 0.$$

Then there exist $C, \gamma > 0$ such that, with $d := |A| + |B| - 2$, for every $P \in \mathcal{K}$, every $r > 0$, and every $q_0 \in \mathcal{G}_{C+r}(\bar{x})$,

$$\mathbb{P}^P_{q_0} \left(q_t \in \mathcal{G}_C(\bar{x}) \textit{ for every } t \geq 0 \textit{ and } q_t \to (\delta_{\bar{a}}, \delta_{\bar{b}}) \right) \geq 1 - de^{-\gamma r}.$$

Apply Proposition A.1 to the singleton $\mathcal{K} = \{P\}$. Strict coordinate-wise KL minimality gives the required positive-gap condition. The proposition therefore supplies $C, \gamma > 0$ such that, with $d := |A| + |B| - 2$, for every $r > 0$ and every $q_0 \in \mathcal{G}_{C+r}(\bar{x})$,

$$\mathbb{P}^* \left(q_t \in \mathcal{G}_C(\bar{x}) \text{ for every } t \geq 0 \text{ and } q_t \to (\delta_{\bar{a}}, \delta_{\bar{b}}) \right) \geq 1 - de^{-\gamma r}.$$

Dropping the nonexit requirement gives the conclusion of part (ii). ■

Proof of Corollary 1

Suppose (i) holds. Let $C, \gamma > 0$ be supplied by Theorem 1(ii), and write $d := |A| + |B| - 2$. For any $\varepsilon \in (0, 1)$, choose $r_\varepsilon > 0$ such that $de^{-\gamma r_\varepsilon} \leq \varepsilon$, and set $C_\varepsilon := C + r_\varepsilon$. Then every $q_0 \in \mathcal{G}_{C_\varepsilon}(\bar{x})$ satisfies

$$\mathbb{P}^* \left(q_t \to (\delta_{\bar{a}}, \delta_{\bar{b}}) \right) \geq 1 - \varepsilon,$$

so (ii) holds.

Suppose (ii) holds. Set $C := C_{1/2}$, and define

$$\alpha_0(\bar{a}) := \frac{e^C}{e^C + |A| - 1}, \quad \alpha_0(a) := \frac{1}{e^C + |A| - 1} \quad \text{for } a \neq \bar{a},$$
$$\beta_0(\bar{b}) := \frac{e^C}{e^C + |B| - 1}, \quad \beta_0(b) := \frac{1}{e^C + |B| - 1} \quad \text{for } b \neq \bar{b}.$$

Then $q_0 = (\alpha_0, \beta_0) \in Q \cap \mathcal{G}_C(\bar{x})$, so

$$\mathbb{P}^* \left(q_t \to (\delta_{\bar{a}}, \delta_{\bar{b}}) \right) \geq \frac{1}{2} > 0.$$

Thus (iii) holds. Finally, suppose (iii) holds. Theorem 1(i) implies that $\bar{x}$ is a weak coordinate-wise KL minimum. The absence of one-coordinate KL ties makes every defining inequality strict, so $\bar{x}$ is a strict coordinate-wise KL minimum. Hence (i) holds. ∎

Proof of Theorem 2

Define

$$E_x := \{q_t \to (\delta_a, \delta_b)\}, \qquad h_x^P(q) := \mathbb{P}_q^P(E_x).$$

The recursive process is a time-homogeneous Markov chain driven by independent signals, so

$$h_x^P(q_t) = \mathbb{P}_{q_0}^P(E_x \mid \mathcal{F}_t).$$

Since $E_x \in \mathcal{F}_\infty := \sigma(\bigcup_{t \geq 0} \mathcal{F}_t)$, Lévy's upward theorem gives[11]

$$h_x^P(q_t) \to \mathbf{1}_{E_x} \qquad \mathbb{P}_{q_0}^P\text{-a.s.} \tag{18}$$

Apply Proposition A.1 to the singleton $\{P\}$. There exist $C, \gamma > 0$ such that, with $d := |A| + |B| - 2$,

$$q' \in \mathcal{G}_{C+r}(x) \quad \Longrightarrow \quad \mathbb{P}_{q'}^P(E_x) \geq 1 - de^{-\gamma r}.$$

Choose $r > 0$ such that $\theta_x := 1 - de^{-\gamma r} > 0$. Fix $\varepsilon \in (0, 1)$. Finite accessibility supplies a word $w = (y_1, \ldots, y_T)$ such that $U_w^P(q) \in \mathcal{G}_{C+r}(x)$ whenever $m_x(q) \geq \varepsilon$. Since the true likelihood has full support, the word has probability

$$p_w^P := \prod_{s=1}^{T} P_{a^* b^*}(y_s) > 0.$$

[11] Lévy's upward theorem states that if $(\mathcal{F}_t)_{t \geq 0}$ is an increasing sequence of sigma-algebras, $\mathcal{F}_\infty := \sigma(\bigcup_{t \geq 0} \mathcal{F}_t)$, and X is integrable, then $\mathbb{E}[X \mid \mathcal{F}_t] \to \mathbb{E}[X \mid \mathcal{F}_\infty]$ almost surely. In particular, if $E \in \mathcal{F}_\infty$ and $X = \mathbf{1}_E$, then $\mathbb{E}[\mathbf{1}_E \mid \mathcal{F}_t] \to \mathbf{1}_E$ almost surely.

The Markov property therefore gives,[12] for every $q \in Q$ satisfying $m_x(q) \geq \varepsilon$,

$$h_x^P(q) \geq \mathbb{P}_q^P\big((Y_1, \ldots, Y_T) = w\big) h_x^P\big(U_w^P(q)\big) = p_w^P h_x^P\big(U_w^P(q)\big) \geq p_w^P \theta_x =: c_{x,\varepsilon} > 0.$$

This proves the asserted uniform lower bound.

Now fix $q_0 \in Q$. Since q_0 has full support, $m_x(q_0) > 0$. Taking $\varepsilon = m_x(q_0)/2$ in the preceding lower bound gives

$$\mathbb{P}_{q_0}^P(E_x) = h_x^P(q_0) \geq c_{x, m_x(q_0)/2} > 0.$$

Finally, on E_x^c, (18) gives $h_x^P(q_t) \to 0$. If $m_x(q_t) \not\to 0$, then $\limsup_{t\to\infty} m_x(q_t) > 0$. Choose $n \geq 2$ such that

$$\frac{1}{n} < \limsup_{t\to\infty} m_x(q_t).$$

Then $m_x(q_t) \geq 1/n$ along infinitely many dates. At every such date, $h_x^P(q_t) \geq c_{x,1/n} > 0$, contradicting $h_x^P(q_t) \to 0$. Hence $m_x(q_t) \to 0$ almost surely on E_x^c. ■

Proof of Theorem 3

For $y \in \{0, 1\}$ and an integer $n \geq 0$, let y^n denote n consecutive occurrences of y, with y^0 the empty word. Recall that $x^* = (L, s)$, and $\bar{x} = (H, g)$, and

$$P_{Hg}^0(1) = \frac{3}{5}, \qquad P_{Hs}^0(1) = \frac{1}{10}, \qquad P_{Lg}^0(1) = \frac{9}{10}, \qquad P_{Ls}^0(1) = \frac{1}{2}.$$

We use the following uniform accessibility result.

Proposition A.2. *There exists an open neighborhood $\mathcal{O}_0$ of P^0, with compact closure contained in $\mathcal{P}$, such that the following holds for every $\varepsilon \in (0,1)$ and every $C > 0$.*

(i) *There exist an integer $n_{\bar{x}}$ and $\rho_{\bar{x}} > 0$ such that, for every $P \in \mathcal{O}_0$ and every $q = (\alpha, \beta) \in Q$ with $\alpha(H)\beta(g) \geq \varepsilon$, defining $w_{\bar{x}} := 1^{n_{\bar{x}}} 0^{\lfloor n_{\bar{x}}/3 \rfloor}$ and $T_{\bar{x}} := n_{\bar{x}} + \lfloor n_{\bar{x}}/3 \rfloor$,*[13]

$$U_{w_{\bar{x}}}^P(q) \in \mathcal{G}_C(\bar{x}), \qquad \mathbb{P}^P((Y_1, \ldots, Y_{T_{\bar{x}}}) = w_{\bar{x}}) \geq \rho_{\bar{x}}.$$

(ii) *There exist an integer n_{x^*} and $\rho_{x^*} > 0$ such that, for every $P \in \mathcal{O}_0$ and every $q =$*

[12] The Markov property means that, for every measurable set A of future paths,

$$\mathbb{P}_q^P\big((q_{t+s})_{s\geq 0} \in A \mid \mathcal{F}_t\big) = \mathbb{P}_{q_t}^P\big((q_s)_{s\geq 0} \in A\big) \qquad \text{a.s.}$$

Here this follows because $q_{t+1} = U_{Y_{t+1}}^P(q_t)$ and, conditional on the true state, future signals are independent of $\mathcal{F}_t$.

[13] For a real number z, $\lfloor z \rfloor$ is the greatest integer not exceeding z, and $\lceil z \rceil$ is the least integer not smaller than z.

$(\alpha, \beta) \in Q$ with $\alpha(L)\beta(s) \geq \varepsilon$, defining $w_{x^*} := 0^{n_{x^*}} 1^{\lfloor 2n_{x^*}/5 \rfloor}$ and $T_{x^*} := n_{x^*} + \lfloor 2n_{x^*}/5 \rfloor$,

$$U^P_{w_{x^*}}(q) \in \mathcal{G}_C(x^*), \qquad \mathbb{P}^P((Y_1, \ldots, Y_{T_{x^*}}) = w_{x^*}) \geq \rho_{x^*}.$$

The integers and probability bounds may depend on ε and C, but not on P or the initial record.

We also use the following no-switching result.

Lemma A.2. *Fix $P \in \mathcal{P}$. Let $A = \{a_0, a_1\}$, $B = \{b_0, b_1\}$, and*

$$\mathcal{V}_{\text{off}} := \{(\delta_{a_0}, \delta_{b_1}), (\delta_{a_1}, \delta_{b_0})\}.$$

Let $(y_t)_{t \geq 1}$ be any sequence in Y, and let $(q_t)_{t \geq 0} \subseteq \overline{Q}$ satisfy $q_{t+1} = U^P_{y_{t+1}}(q_t)$ for every $t \geq 0$. If $\operatorname{dist}(q_t, \mathcal{V}_{\text{off}}) \to 0$, then q_t converges to one element of $\mathcal{V}_{\text{off}}$.

Step 1 (A common neighborhood and uniform KL gaps). Let $\mathcal{O}_0$ be supplied by Proposition A.2. At P^0, the true state (L, s) is identified and (H, g) is a strict coordinate-wise KL minimum. Continuity of $K^P(a, b)$ in P allows us to choose an open neighborhood $\mathcal{O}$ of P^0 such that $\overline{\mathcal{O}} \subset \mathcal{O}_0$, $\overline{\mathcal{O}}$ is compact, and

$$\min_{P \in \overline{\mathcal{O}}} \min\{K^P(H, g), K^P(H, s), K^P(L, g)\} > 0, \tag{19}$$

$$\min_{P \in \overline{\mathcal{O}}} \min\{K^P(L, g) - K^P(H, g),\ K^P(H, s) - K^P(H, g)\} > 0. \tag{20}$$

Since $K^P(L, s) = 0$, (19) identifies (L, s) throughout $\overline{\mathcal{O}}$, while (20) makes (H, g) a strict coordinate-wise KL minimum throughout the same set. Identification also makes (L, s) a strict coordinate-wise KL minimum.

Step 2 (Full Bayesian consistency). Fix $P \in \mathcal{O}$ and $q_0 \in Q$. The initial belief $\pi^F_0 = R(q_0)$ has full support, and (L, s) is identified. Proposition 2 gives

$$\pi^F_t \to \delta_{(L,s)} \qquad \mathbb{P}^P\text{-a.s.}$$

Step 3 (Exhaustive diagonal selection). Define

$$E_{x^*} := \{q_t \to (\delta_L, \delta_s)\}, \qquad E_{\bar{x}} := \{q_t \to (\delta_H, \delta_g)\}.$$

Both x^* and $\bar{x}$ are strict coordinate-wise KL minima, and Proposition A.2 implies that both are finitely accessible away from zero product mass. Theorem 2 therefore gives

$$\mathbb{P}^P_{q_0}(E_{x^*}) > 0, \qquad \mathbb{P}^P_{q_0}(E_{\bar{x}}) > 0.$$

On $E^c_{x^*} \cap E^c_{\bar{x}}$, the same theorem gives

$$m_{x^*}(q_t) \to 0, \qquad m_{\bar{x}}(q_t) \to 0 \qquad \mathbb{P}^P_{q_0}\text{-a.s.}$$

Write $p_t := \alpha_t(L)$ and $r_t := \beta_t(s)$. Then

$$p_t r_t \to 0, \qquad (1 - p_t)(1 - r_t) \to 0. \tag{21}$$

Every subsequential limit $(p, r) \in [0, 1]^2$ satisfying (21) must equal either $(1, 0)$ or $(0, 1)$. These points correspond to the off-diagonal pure records

$$\mathcal{V}_{\text{off}} := \{(\delta_L, \delta_g), (\delta_H, \delta_s)\}.$$

Hence $\text{dist}(q_t, \mathcal{V}_{\text{off}}) \to 0$ almost surely on $E^c_{x^*} \cap E^c_{\bar{x}}$. Otherwise, compactness of $\overline{Q}$ would yield a subsequential limit outside $\mathcal{V}_{\text{off}}$, contradicting (21). Lemma A.2, applied with $a_0 = L$, $a_1 = H$, $b_0 = s$, and $b_1 = g$, then implies that the record converges to one of the two off-diagonal pure records.

Neither off-diagonal state is a weak coordinate-wise KL minimum. Identification gives $K^P(L, s) < K^P(L, g)$, so (L, g) loses the comparison obtained by changing only its second coordinate. Similarly, (20) gives $K^P(H, g) < K^P(H, s)$, so (H, s) also loses a one-coordinate comparison. Theorem 1(i), applied to the fixed likelihood system P, assigns probability zero to convergence to either off-diagonal pure record. Therefore, $\mathbb{P}^P_{q_0}(E^c_{x^*} \cap E^c_{\bar{x}}) = 0$, and hence $\mathbb{P}^P_{q_0}(E_{x^*} \cup E_{\bar{x}}) = 1$.

Step 4 (A uniform false-convergence lower bound). By (20), Proposition A.1, applied to $\overline{\mathcal{O}}$ and $\bar{x} = (H, g)$, gives constants $C_{\bar{x}}, \gamma_{\bar{x}} > 0$ such that, for every $P \in \overline{\mathcal{O}}$, every $r > 0$, and every $q \in \mathcal{G}_{C_{\bar{x}}+r}(\bar{x})$,

$$\mathbb{P}^P_q\left(q_t \to (\delta_H, \delta_g)\right) \geq 1 - 2e^{-\gamma_{\bar{x}} r}. \tag{22}$$

Choose $r_0 > 0$ such that $1 - 2e^{-\gamma_{\bar{x}} r_0} \geq 1/2$. Fix $\zeta > 0$. If $q_0 = (\alpha, \beta) \in Q_\zeta$, then $\alpha(H), \alpha(L), \beta(g), \beta(s) \geq (1 + e^\zeta)^{-1}$, so

$$m_{\bar{x}}(q_0) = \alpha(H)\beta(g) \geq \varepsilon_\zeta := \frac{1}{(1 + e^\zeta)^2}.$$

Apply Proposition A.2(i) with $\varepsilon = \varepsilon_\zeta$ and $C = C_{\bar{x}} + r_0$. There exist an integer n_ζ and $\rho_\zeta > 0$, independent of $P \in \mathcal{O}$ and $q_0 \in Q_\zeta$, such that, with

$$w_\zeta := 1^{n_\zeta} 0^{\lfloor n_\zeta/3 \rfloor}, \qquad T_\zeta := n_\zeta + \lfloor n_\zeta/3 \rfloor,$$

we have $U^P_{w_\zeta}(q_0) \in \mathcal{G}_{C_{\bar{x}}+r_0}(\bar{x})$ and

$$\mathbb{P}^P((Y_1, \ldots, Y_{T_\zeta}) = w_\zeta) \geq \rho_\zeta.$$

Let $A_\zeta := \{(Y_1, \ldots, Y_{T_\zeta}) = w_\zeta\}$. On A_ζ, the Markov property and (22) give

$$\mathbb{P}^P_{q_0}(E_{\bar{x}} \mid \mathcal{F}_{T_\zeta}) = \mathbb{P}^P_{q_{T_\zeta}}(E_{\bar{x}}) \geq \frac{1}{2}.$$

Therefore,

$$\mathbb{P}^P_{q_0}(E_{\bar{x}}) \geq \mathbb{E}^P\left[\mathbf{1}_{A_\zeta} \mathbb{P}^P_{q_0}(E_{\bar{x}} \mid \mathcal{F}_{T_\zeta})\right] \geq \frac{1}{2}\mathbb{P}^P(A_\zeta) \geq \frac{\rho_\zeta}{2}.$$

Setting $\eta_\zeta := \rho_\zeta/2 > 0$ proves $\inf_{P\in\mathcal{O}} \inf_{q_0 \in Q_\zeta} \mathbb{P}^P_{q_0}\left(q_t \to (\delta_H, \delta_g)\right) \geq \eta_\zeta$. ■

Proof of Theorem 4

We prove (i) ⇒ (ii) ⇒ (iii) ⇒ (i).

(i) **implies** (ii). Suppose $P_{ab}(y) = f^y_A(a) f^y_B(b)$. Fix $q = (\alpha, \beta) \in Q$ and $y \in Y$. Bayes' rule and factorization of the denominator give

$$\mathsf{B}_y(R(q))(a,b) = \frac{\alpha(a) f^y_A(a)}{\sum_{a' \in A} \alpha(a') f^y_A(a')} \frac{\beta(b) f^y_B(b)}{\sum_{b' \in B} \beta(b') f^y_B(b')}.$$

Thus the posterior is a product belief.

(ii) **implies** (iii). Fix a finite signal history $(y_1, \ldots, y_T)$. We prove by induction that $R(q_t) = \pi^F_t$ for $t = 0, \ldots, T$. The equality holds at $t = 0$. If it holds at $t < T$, then

$$\pi^F_{t+1} = \mathsf{B}_{y_{t+1}}(\pi^F_t) = \mathsf{B}_{y_{t+1}}(R(q_t)).$$

By (ii), this posterior is a product belief, so

$$\pi^F_{t+1} = R\left(\mathsf{M}\big(\mathsf{B}_{y_{t+1}}(R(q_t))\big)\right) = R(q_{t+1}).$$

This completes the induction.

(iii) **implies** (i). Fix $y \in Y$, and let $q^u = (\alpha^u, \beta^u)$ be the uniform marginal record, with $\alpha^u(a) = 1/|A|$ and $\beta^u(b) = 1/|B|$. Initialize the full Bayesian learner from $R(q^u)$ and consider the one-signal history y. Then

$$\pi_1^F(a,b) = \frac{P_{ab}(y)}{\sum_{a' \in A} \sum_{b' \in B} P_{a'b'}(y)}.$$

By (iii), π_1^F is a product belief. Hence, for every $a, a' \in A$ and $b, b' \in B$,

$$\pi_1^F(a,b)\pi_1^F(a',b') = \pi_1^F(a,b')\pi_1^F(a',b).$$

The common positive normalizing constant cancels, yielding

$$P_{ab}(y)P_{a'b'}(y) = P_{ab'}(y)P_{a'b}(y) \qquad \text{for every } a, a' \in A,\ b, b' \in B. \tag{23}$$

Fix reference coordinates $a_0 \in A$ and $b_0 \in B$, and define

$$f_A^y(a) := P_{ab_0}(y), \qquad f_B^y(b) := \frac{P_{a_0 b}(y)}{P_{a_0 b_0}(y)}.$$

Full support makes these functions strictly positive. Applying (23) with $a' = a_0$ and $b' = b_0$ gives $P_{ab}(y) = f_A^y(a) f_B^y(b)$. Since y was arbitrary, the likelihood system has no coordinate interaction.

Finally, under this factorization,

$$K(a,b) = \sum_{y \in Y} P^*(y) \log P^*(y) - \sum_{y \in Y} P^*(y) \log f_A^y(a) - \sum_{y \in Y} P^*(y) \log f_B^y(b) = c + \kappa_A(a) + \kappa_B(b).$$

If $(\bar{a}, \bar{b})$ is a weak coordinate-wise KL minimum, then $\bar{a}$ minimizes κ_A and $\bar{b}$ minimizes κ_B, so $K(\bar{a}, \bar{b}) \leq K(a,b)$ for every $(a,b) \in X$. Thus every weak coordinate-wise KL minimum is a global KL minimizer. Under identification, x^* is the unique global KL minimizer and therefore the unique weak coordinate-wise KL minimum. ∎

Proof of Corollary 2

Let $p_A := \pi_A(a_1)$, $p_B := \pi_B(b_1)$, and $z := \pi(a_1, b_1)$. The remaining cell probabilities are $p_A - z$, $p_B - z$, and $1 - p_A - p_B + z$,[14] so full support is equivalent to

$$\max\{0, p_A + p_B - 1\} < z < \min\{p_A, p_B\}. \tag{24}$$

[14] For example, $p_A = \pi(a_1, b_0) + \pi(a_1, b_1)$, so $\pi(a_1, b_0) = p_A - z$.

For fixed marginals, define

$$f_{p_A,p_B}(z) := \log \frac{z(1-p_A-p_B+z)}{(p_A-z)(p_B-z)}.$$

On the interval in (24),

$$f'_{p_A,p_B}(z) = \frac{1}{z} + \frac{1}{1-p_A-p_B+z} + \frac{1}{p_A-z} + \frac{1}{p_B-z} > 0.$$

Moreover, $f_{p_A,p_B}(z) \to -\infty$ at the lower endpoint of (24) and $f_{p_A,p_B}(z) \to +\infty$ at the upper endpoint. Thus, for fixed marginals, each value of $\ell(\pi)$ determines a unique z, and the marginals determine the other three cells. Hence $(\mathsf{M}(\pi), \ell(\pi))$ uniquely determines π.

Substituting Bayes' rule into the posterior cross-ratio gives

$$\exp\big(\ell(\mathsf{B}_y(\pi))\big) = \frac{\pi(a_1,b_1)\pi(a_0,b_0)}{\pi(a_1,b_0)\pi(a_0,b_1)} \frac{P_{a_1b_1}(y)P_{a_0b_0}(y)}{P_{a_1b_0}(y)P_{a_0b_1}(y)} = \exp\big(\ell(\pi)+\lambda_y\big).$$

The common posterior normalizing factor cancels, so $\ell(\mathsf{B}_y(\pi)) = \ell(\pi)+\lambda_y$. Since $R(\mathsf{M}(\pi))$ is a product belief, $\ell(R(\mathsf{M}(\pi))) = 0$. If $\pi_0^F = R(q_0)$, then $\ell(\pi_0^F) = 0$, and repeated application of the update identity gives

$$\ell(\pi_t^F) = \sum_{\tau=1}^{t} \lambda_{Y_\tau}.$$

The uniqueness argument shows that $(\mathsf{M}(\pi_t^F), \ell(\pi_t^F))$ determines π_t^F at every date.

Finally, $\lambda_y = 0$ if and only if

$$P_{a_1b_1}(y)P_{a_0b_0}(y) = P_{a_1b_0}(y)P_{a_0b_1}(y).$$

For a fixed signal realization in a full-support 2×2 environment, this equality is equivalent to the factorization in (10). Hence $\lambda_y = 0$ for every $y \in Y$ if and only if the likelihood system has no coordinate interaction. Theorem 4 then implies that the marginal record alone reproduces full Bayesian learning from every product prior exactly under this condition. ■

A.2 Auxiliary results and proofs

Proof of Lemma A.1

Let $M_t := \sum_{s=1}^t \xi_s$. Azuma–Hoeffding's inequality gives[15], for every $\varepsilon > 0$,

$$\mathbb{P}(|M_t| > \varepsilon t) \le 2\exp\left(-\frac{\varepsilon^2 t}{2L^2}\right).$$

The right-hand side is summable in t, so the first Borel–Cantelli lemma implies that $|M_t| > \varepsilon t$ occurs only finitely often almost surely.[16] Applying this conclusion to $\varepsilon = 1/n$, $n \ge 1$, gives $M_t/t \to 0$ almost surely. ∎

Lemma A.3. *Let $\mathcal{K} \subset \mathcal{P}$ be compact.*

(i) *For every $y \in Y$, the map $(P,q) \mapsto U_y^P(q)$ is continuous on $\mathcal{K} \times \overline{Q}$.*

(ii) *Every pure record is fixed:*

$$U_y^P(\delta_a, \delta_b) = (\delta_a, \delta_b)$$

for every $P \in \mathcal{K}$, $y \in Y$, and $(a,b) \in X$.

(iii) *For every $y \in Y$, every distinct $a, a' \in A$, and every distinct $b, b' \in B$, the maps*

$$(P,\beta) \mapsto \log \frac{\widetilde{P}_A^P(y \mid a; \beta)}{\widetilde{P}_A^P(y \mid a'; \beta)}, \qquad (P,\alpha) \mapsto \log \frac{\widetilde{P}_B^P(y \mid b; \alpha)}{\widetilde{P}_B^P(y \mid b'; \alpha)}$$

are jointly continuous and uniformly bounded on $\mathcal{K} \times \overline{\Delta}(B)$ and $\mathcal{K} \times \overline{\Delta}(A)$, respectively.

(iv) *For a fixed candidate $\bar{x} = (\bar{a}, \bar{b})$, define*

$$\Phi_a^P(\beta) := \sum_{y \in Y} P_{a^*b^*}(y) \log \frac{\widetilde{P}_A^P(y \mid \bar{a}; \beta)}{\widetilde{P}_A^P(y \mid a; \beta)}, \qquad a \ne \bar{a},$$

$$\Psi_b^P(\alpha) := \sum_{y \in Y} P_{a^*b^*}(y) \log \frac{\widetilde{P}_B^P(y \mid \bar{b}; \alpha)}{\widetilde{P}_B^P(y \mid b; \alpha)}, \qquad b \ne \bar{b}.$$

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[15] Azuma–Hoeffding's inequality states that if $(M_t, \mathcal{F}_t)_{t \ge 0}$ is a martingale and there are constants $c_1, \dots, c_t$ such that $|M_s - M_{s-1}| \le c_s$ almost surely for $s = 1, \dots, t$, then, for every $x > 0$,

$$\mathbb{P}\left(|M_t - M_0| \ge x\right) \le 2\exp\left\{-\frac{x^2}{2\sum_{s=1}^t c_s^2}\right\}.$$

Thus, if $M_0 = 0$ and every increment is bounded in absolute value by L, then $\mathbb{P}(|M_t| \ge x) \le 2\exp\left\{-\frac{x^2}{2tL^2}\right\}$.

[16] The first Borel–Cantelli lemma states that for any sequence of events $(A_t)_{t \ge 1}$, if $\sum_{t=1}^\infty \mathbb{P}(A_t) < \infty$, then $\mathbb{P}(A_t \text{ infinitely often}) = 0$. No independence assumption is required.

These functions are jointly continuous in (P,β) and (P,α), respectively.

Proof of Lemma A.3

Compactness within the full-support likelihood space gives

$$\underline{p}_{\mathcal{K}} := \min_{P\in\mathcal{K}} \min_{(a,b)\in X} \min_{y\in Y} P_{ab}(y) > 0.$$

Every effective likelihood lies in $[\underline{p}_{\mathcal{K}}, 1]$, and every Bayesian normalizing denominator is at least $\underline{p}_{\mathcal{K}}$. The update maps are therefore ratios of continuous functions with uniformly positive denominators, proving (i). If $q = (\delta_a, \delta_b)$, then $R(q) = \delta_{(a,b)}$, and Bayesian updating preserves this degenerate belief; taking marginals proves (ii). The same lower bound implies that every effective log-likelihood ratio is bounded in absolute value by $-\log \underline{p}_{\mathcal{K}}$, while joint continuity follows from the effective-likelihood formulas, proving (iii). The drift functions are finite sums of products of continuous functions, proving (iv). ■

Lemma A.4. *Let $Z_t^1, \ldots, Z_t^d$ be real-valued processes adapted to $(\mathcal{F}_t)$, and write*

$$\Delta Z_{t+1}^j := Z_{t+1}^j - Z_t^j.$$

Fix $C \in \mathbb{R}$, and define $\tau := \inf\{t \geq 0 : \min_{1\leq j\leq d} Z_t^j < C\}$. Suppose there exist $L < \infty$ and $\kappa > 0$ such that, for every j, $|\Delta Z_{t+1}^j| \leq L$ almost surely and $\mathbb{E}[\Delta Z_{t+1}^j \mid \mathcal{F}_t] \geq \kappa$ on $\{t < \tau\}$. Then there exists $\gamma > 0$, depending only on L and κ, such that, for every $r > 0$,

$$\min_{1\leq j\leq d} Z_0^j \geq C + r \quad \Longrightarrow \quad \mathbb{P}(\tau < \infty) \leq d e^{-\gamma r}.$$

On $\{\tau = \infty\}$, $\liminf_{t\to\infty} Z_t^j / t \geq \kappa$ for every j.

Proof of Lemma A.4

Choose $\theta > 0$ sufficiently small that $\theta L^2 e^{\theta L}/2 \leq \kappa$. For $|\Delta Z| \leq L$, Taylor's formula gives

$$e^{-\theta \Delta Z} \leq 1 - \theta \Delta Z + \frac{\theta^2 L^2 e^{\theta L}}{2}.$$

Hence, on $\{t < \tau\}$,

$$\mathbb{E}[e^{-\theta \Delta Z_{t+1}^j} \mid \mathcal{F}_t] \leq 1 - \theta\kappa + \frac{\theta^2 L^2 e^{\theta L}}{2} \leq 1.$$

Define $S_t^j := \exp\{-\theta(Z_{\min\{t,\tau\}}^j - Z_0^j)\}$. On $\{t < \tau\}$, $S_{t+1}^j / S_t^j = e^{-\theta \Delta Z_{t+1}^j}$, while on $\{t \geq \tau\}$ this ratio equals one. Hence the preceding conditional-moment bound implies $\mathbb{E}[S_{t+1}^j \mid \mathcal{F}_t] \leq S_t^j$.

Thus $(S_t^j)_{t\geq 0}$ is a nonnegative supermartingale. If $\min_j Z_0^j \geq C + r$, then

$$\{\tau < \infty,\ Z_\tau^j < C\} \subseteq \left\{\sup_{t\geq 0} S_t^j \geq e^{\theta r}\right\}.$$

Ville's inequality gives[17] $\mathbb{P}(\tau < \infty, Z_\tau^j < C) \leq e^{-\theta r}$, and a union bound over j gives $\mathbb{P}(\tau < \infty) \leq de^{-\theta r}$. Set $\gamma := \theta$.

For the growth conclusion, write

$$Z_t^j = Z_0^j + \sum_{s=0}^{t-1} \mathbb{E}[\Delta Z_{s+1}^j \mid \mathcal{F}_s] + M_t^j, \qquad M_t^j := \sum_{s=0}^{t-1} \left(\Delta Z_{s+1}^j - \mathbb{E}[\Delta Z_{s+1}^j \mid \mathcal{F}_s]\right).$$

The martingale increments are bounded by $2L$, so Lemma A.1 gives $M_t^j/t \to 0$. On $\{\tau = \infty\}$, every conditional drift is at least κ, and dividing by t yields $\liminf_{t\to\infty} Z_t^j/t \geq \kappa$. ■

Proof of Proposition A.1

For $a \neq \bar{a}$ and $b \neq \bar{b}$, define

$$Z_t^a := \log \frac{\alpha_t(\bar{a})}{\alpha_t(a)}, \qquad W_t^b := \log \frac{\beta_t(\bar{b})}{\beta_t(b)}.$$

At $\beta = \delta_{\bar{b}}$ and $\alpha = \delta_{\bar{a}}$,

$$\Phi_a^P(\delta_{\bar{b}}) = K^P(a, \bar{b}) - K^P(\bar{a}, \bar{b}), \qquad \Psi_b^P(\delta_{\bar{a}}) = K^P(\bar{a}, b) - K^P(\bar{a}, \bar{b}).$$

By the assumed uniform gap and Lemma A.3, there exist neighborhoods N_A of $\delta_{\bar{a}}$ and N_B of $\delta_{\bar{b}}$ such that

$$\Phi_a^P(\beta) \geq \frac{\kappa_0}{2}, \qquad \Psi_b^P(\alpha) \geq \frac{\kappa_0}{2}$$

for every $P \in \mathcal{K}$, $\beta \in N_B$, $\alpha \in N_A$, $a \neq \bar{a}$, and $b \neq \bar{b}$. If $\min_{a\neq\bar{a}} \log(\alpha(\bar{a})/\alpha(a)) \geq C$, then

$$\alpha(\bar{a}) \geq \frac{1}{1 + (|A| - 1)e^{-C}},$$

and a symmetric bound holds for $\beta(\bar{b})$. Hence C can be chosen so that $\mathcal{G}_C(\bar{x}) \subseteq N_A \times N_B$.

[17] Ville's inequality states that if $(S_t, \mathcal{F}_t)_{t\geq 0}$ is a nonnegative supermartingale, then, for every $a > 0$,

$$\mathbb{P}\left(\sup_{t\geq 0} S_t \geq a\right) \leq \frac{\mathbb{E}[S_0]}{a}.$$

Here $S_0^j = 1$ and $a = e^{\theta r}$.

Define $\tau := \inf\{t \geq 0 : q_t \notin \mathcal{G}_C(\bar{x})\}$. Before τ, every process Z_t^a and W_t^b has conditional drift at least $\kappa_0/2$, and Lemma A.3 gives a common finite bound L on their increments. Applying Lemma A.4 to the $d := |A| + |B| - 2$ log-odds processes gives $\gamma > 0$ such that, for every $P \in \mathcal{K}$, every $r > 0$, and every $q_0 \in \mathcal{G}_{C+r}(\bar{x})$,

$$\mathbb{P}_{q_0}^P(\tau < \infty) \leq d e^{-\gamma r}.$$

On $\{\tau = \infty\}$, $Z_t^a \to +\infty$ and $W_t^b \to +\infty$ for every alternative coordinate. Therefore,

$$\alpha_t(\bar{a}) = \frac{1}{1 + \sum_{a \neq \bar{a}} e^{-Z_t^a}} \to 1, \qquad \beta_t(\bar{b}) = \frac{1}{1 + \sum_{b \neq \bar{b}} e^{-W_t^b}} \to 1,$$

so $q_t \to (\delta_{\bar{a}}, \delta_{\bar{b}})$. Combining this conclusion with the exit bound proves the proposition. ■

Proof of Proposition A.2

For a binary record $q = (\alpha, \beta)$, write

$$u := \log \frac{\alpha(H)}{\alpha(L)}, \qquad v := \log \frac{\beta(g)}{\beta(s)}, \qquad \Lambda(z) := \frac{e^z}{1 + e^z}.$$

Then

$$u_{t+1} = u_t + g_u^P(Y_{t+1}, v_t), \qquad v_{t+1} = v_t + g_v^P(Y_{t+1}, u_t),$$

where

$$g_u^P(y, v) := \log \frac{\Lambda(v) P_{Hg}(y) + (1 - \Lambda(v)) P_{Hs}(y)}{\Lambda(v) P_{Lg}(y) + (1 - \Lambda(v)) P_{Ls}(y)},$$
$$g_v^P(y, u) := \log \frac{\Lambda(u) P_{Hg}(y) + (1 - \Lambda(u)) P_{Lg}(y)}{\Lambda(u) P_{Hs}(y) + (1 - \Lambda(u)) P_{Ls}(y)}.$$

Step 1 (Uniform bounds for steering toward (H, g)). At P^0, putting $p = \Lambda(u)$ gives

$$e^{g_v^{P^0}(1,u)} = \frac{9 - 3p}{5 - 4p}, \qquad e^{g_v^{P^0}(0,u)} = \frac{1 + 3p}{5 + 4p}.$$

Both ratios are increasing in p, so

$$\inf_u g_v^{P^0}(1, u) = \log \frac{9}{5}, \qquad \inf_u g_v^{P^0}(0, u) = -\log 5.$$

Moreover,

$$\lim_{v \to +\infty} g_u^{P^0}(1, v) = -\log \frac{3}{2}, \qquad \lim_{v \to +\infty} g_u^{P^0}(0, v) = \log 4.$$

Since $\log\frac{9}{5} > \frac{\log 5}{3}$, and $\frac{\log 4}{3} > \log\frac{3}{2}$, choose positive a_g, e_g, b_H, c_H such that

$$a_g < \log\frac{9}{5}, \quad e_g > \log 5, \quad b_H > \log\frac{3}{2}, \quad c_H < \log 4, \quad a_g > \frac{e_g}{3}, \quad \frac{c_H}{3} > b_H.$$

Because $g_v^P(y,u)$ depends on u only through $\Lambda(u)$, it extends continuously to (P,p) with $p \in [0,1]$. Similarly, $g_u^P(y,v)$ depends on v only through $\Lambda(v)$. Compactness and uniform continuity therefore give an open neighborhood $\mathcal{N}_g$ of P^0 and $V_g \in \mathbb{R}$ such that, uniformly over $P \in \mathcal{N}_g$,

$$\begin{aligned} &g_v^P(1,u) \geq a_g, \qquad g_v^P(0,u) \geq -e_g \quad \text{for every } u \in \mathbb{R}, \\ &g_u^P(1,v) \geq -b_H, \qquad g_u^P(0,v) \geq c_H \quad \text{whenever } v \geq V_g. \end{aligned} \tag{25}$$

Step 2 (Uniform bounds for steering toward (L,s)). Define $\bar{u} := -u = \log(\alpha(L)/\alpha(H))$ and $\bar{v} := -v = \log(\beta(s)/\beta(g))$. At P^0,

$$\inf_{\bar{u}}\{-g_v^{P^0}(0,-\bar{u})\} = \log\frac{9}{4}, \qquad \inf_{\bar{u}}\{-g_v^{P^0}(1,-\bar{u})\} = -\log 6,$$

and

$$\lim_{\bar{v}\to+\infty}\{-g_u^{P^0}(0,-\bar{v})\} = -\log\frac{9}{5}, \qquad \lim_{\bar{v}\to+\infty}\{-g_u^{P^0}(1,-\bar{v})\} = \log 5.$$

Since $\log\frac{9}{4} > \frac{2\log 6}{5}$, and $\frac{2\log 5}{5} > \log\frac{9}{5}$, choose positive a_s, e_s, b_L, c_L such that

$$a_s < \log\frac{9}{4}, \quad e_s > \log 6, \quad b_L > \log\frac{9}{5}, \quad c_L < \log 5, \quad a_s > \frac{2e_s}{5}, \quad \frac{2c_L}{5} > b_L.$$

The same continuity and compactness argument gives an open neighborhood $\mathcal{N}_s$ of P^0 and $V_s \in \mathbb{R}$ such that, uniformly over $P \in \mathcal{N}_s$,

$$\begin{aligned} &-g_v^P(0,-\bar{u}) \geq a_s, \qquad -g_v^P(1,-\bar{u}) \geq -e_s \quad \text{for every } \bar{u} \in \mathbb{R}, \\ &-g_u^P(0,-\bar{v}) \geq -b_L, \qquad -g_u^P(1,-\bar{v}) \geq c_L \quad \text{whenever } \bar{v} \geq V_s. \end{aligned} \tag{26}$$

Choose an open neighborhood $\mathcal{O}_0$ of P^0 such that $\overline{\mathcal{O}_0} \subset \mathcal{N}_g \cap \mathcal{N}_s$ and $\overline{\mathcal{O}_0} \subset \mathcal{P}$. Lemma A.3 gives a finite constant L_{acc} bounding the absolute value of every one-period change in $u, v, \bar{u}, \bar{v}$, uniformly over $P \in \overline{\mathcal{O}_0}$.

Step 3 (The false steering word). Fix $\varepsilon \in (0,1)$ and $C > 0$, and suppose $\alpha_0(H)\beta_0(g) \geq \varepsilon$. Since both factors are at most one, $\alpha_0(H), \beta_0(g) \geq \varepsilon$, and hence $u_0, v_0 \geq \log\varepsilon$. During a first block of n ones, (25) gives $v_t \geq \log\varepsilon + a_g t$. Define

$$N_g(\varepsilon) := \max\left\{0, \left\lceil \frac{V_g - \log\varepsilon}{a_g} \right\rceil\right\}, \qquad D_g(\varepsilon) := L_{\text{acc}} N_g(\varepsilon) - \log\varepsilon.$$

For $n \geq N_g(\varepsilon)$, the unrestricted increment bound before date $N_g(\varepsilon)$ and the bound $g_u^P(1, v) \geq -b_H$ thereafter imply

$$u_n \geq -b_H n - D_g(\varepsilon).$$

Append $m_n := \lfloor n/3 \rfloor$ zeros. For every $0 \leq k \leq m_n$,

$$v_{n+k} \geq \log \varepsilon + a_g n - e_g k \geq \log \varepsilon + \left(a_g - \frac{e_g}{3}\right) n.$$

For all sufficiently large n, this lower bound exceeds V_g throughout the second block. Each zero then increases u by at least c_H, so

$$u_{n+m_n} \geq \left(\frac{c_H}{3} - b_H\right) n - D_g(\varepsilon) - c_H, \qquad v_{n+m_n} \geq \log \varepsilon + \left(a_g - \frac{e_g}{3}\right) n.$$

Both lower bounds tend to $+\infty$. Hence a finite $n_{\bar{x}}$, depending only on ε and C, can be chosen such that

$$U^P_{1^{n_{\bar{x}}} 0^{\lfloor n_{\bar{x}}/3 \rfloor}}(q_0) \in \mathcal{G}_C(H, g),$$

uniformly over $P \in \mathcal{O}_0$ and all admissible initial records.

Step 4 (The true steering word). Suppose $\alpha_0(L)\beta_0(s) \geq \varepsilon$. Then $\bar{u}_0, \bar{v}_0 \geq \log \varepsilon$. During a first block of n zeros, (26) gives $\bar{v}_t \geq \log \varepsilon + a_s t$. Define

$$N_s(\varepsilon) := \max\left\{0, \left\lceil \frac{V_s - \log \varepsilon}{a_s} \right\rceil\right\}, \qquad D_s(\varepsilon) := L_{\text{acc}} N_s(\varepsilon) - \log \varepsilon.$$

For $n \geq N_s(\varepsilon)$, the unrestricted increment bound before date $N_s(\varepsilon)$ and the bound $-g_u^P(0, -\bar{v}) \geq -b_L$ thereafter imply

$$\bar{u}_n \geq -b_L n - D_s(\varepsilon).$$

Append $k_n := \lfloor 2n/5 \rfloor$ ones. For every $0 \leq k \leq k_n$,

$$\bar{v}_{n+k} \geq \log \varepsilon + a_s n - e_s k \geq \log \varepsilon + \left(a_s - \frac{2e_s}{5}\right) n.$$

For all sufficiently large n, this lower bound exceeds V_s throughout the second block. Each one then increases $\bar{u}$ by at least c_L, so

$$\bar{u}_{n+k_n} \geq \left(\frac{2c_L}{5} - b_L\right) n - D_s(\varepsilon) - c_L, \qquad \bar{v}_{n+k_n} \geq \log \varepsilon + \left(a_s - \frac{2e_s}{5}\right) n.$$

Both lower bounds tend to $+\infty$. Hence a finite n_{x^*}, depending only on ε and C, can be chosen such that $U^P_{0^{n_{x^*}} 1^{\lfloor 2n_{x^*}/5 \rfloor}}(q_0) \in \mathcal{G}_C(L, s)$, uniformly over $P \in \mathcal{O}_0$ and all admissible initial records.

Step 5 (Uniform word probabilities). Compactness and full support give

$$\underline{p} := \min_{P \in \overline{\mathcal{O}_0}} \min\{P_{Ls}(0), P_{Ls}(1)\} > 0.$$

Every binary word of length T therefore has probability at least $\underline{p}^T$, uniformly over $P \in \mathcal{O}_0$. We may take $\rho_{\bar{x}} := \underline{p}^{n_{\bar{x}}+\lfloor n_{\bar{x}}/3 \rfloor}$, and $\rho_{x^*} := \underline{p}^{n_{x^*}+\lfloor 2n_{x^*}/5 \rfloor}$. ∎

Proof of Lemma A.2

Write $z^0 := (\delta_{a_0}, \delta_{b_1})$, $z^1 := (\delta_{a_1}, \delta_{b_0})$, and $D := \|z^0 - z^1\| > 0$. By Lemma A.3, each U_y^P is continuous on $\overline{Q}$ and fixes both vertices. Choose $\eta \in (0, D/3)$. Since Y is finite, continuity gives $\rho \in (0, D/3)$ such that

$$\|q - z^i\| < \rho \quad \Longrightarrow \quad \|U_y^P(q) - z^i\| < \eta \tag{27}$$

for every $i \in \{0, 1\}$ and $y \in Y$.

Suppose $\operatorname{dist}(q_t, \mathcal{V}_{\text{off}}) \to 0$. There exists T such that, for every $t \geq T$, q_t lies within ρ of exactly one off-diagonal vertex. If q_t lies within ρ of z^i, then (27) gives $\|q_{t+1} - z^i\| < \eta$. By the choice of T, q_{t+1} also lies within ρ of one of the two off-diagonal vertices. It cannot lie within ρ of z^{1-i}, because the triangle inequality would imply

$$D < \eta + \rho < \frac{2D}{3}.$$

Hence the nearby vertex remains z^i. By induction, the nearby vertex cannot change after date T. Since $\operatorname{dist}(q_t, \mathcal{V}_{\text{off}}) \to 0$, the record converges to that fixed vertex. ∎

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